

PREMIUM BASIC SCHOOL MANAGEMENT SYSTEM

Complete Enterprise Specification

INTEGRATED EDITION

Volumes I-IV
Chapters 1-55

Nursery - Kindergarten - Primary - Junior High School
Version 1.0 | July 2026

Master Volume Index

This integrated edition combines the four approved PBSMS volumes into one continuous master document. Each volume retains its own cover, document control, table of contents, chapter numbering and page numbering.

Volume	Title	Chapter Range	Original Pages
I	Enterprise Foundation and Academic Architecture	1-21	52
II	Functional Modules and Processing Engines	22-31	27
III	Technical Architecture and Implementation Blueprint	32-45	25
IV	Codex Implementation Blueprint	46-55	24

Document Scope

PBSMS is a configurable, role-based and integrated school-management platform for Nursery through Junior High School. It uses Ghana cedi (GHS) as the default currency while keeping institutional settings configurable.

Scope Boundary

Senior High School, universities, semesters, credit hours, GPA and CGPA are outside the approved scope.

Implementation Principle

Work proceeds as a controlled upgrade of the existing Premium Grading System. Existing functionality and historical data must be inspected, preserved and verified throughout implementation.

Navigation

Use the PDF bookmark panel to move directly between volumes. Each volume also contains its own detailed table of contents.

PREMIUM BASIC SCHOOL MANAGEMENT SYSTEM

Enterprise Software Requirements Specification (SRS)

Nursery - Kindergarten - Primary - Junior High School (JHS)

Document	Value
Version	1.0
Status	Consolidated Baseline
Country Context	Ghana
Default Currency	GHS
Default Time Zone	Africa/Accra

This volume consolidates the approved SRS material through Chapter 21. It excludes Senior High School and tertiary education concepts.

Publication Information

Item	Specification
Project	Premium Basic School Management System (PBSMS)
Document Type	Enterprise Software Requirements Specification
Volume	I - Enterprise Foundation and Academic Architecture
Education Scope	Nursery to JHS 3
Version	1.0
Status	Consolidated baseline for review and phased implementation
Source	Curated and reconciled from the project design conversation
Exclusions	SHS, university, semester, credit-hour, GPA/CGPA and degree-programme models

Revision History

Version	Description
0.x	Working conversation drafts and modular prompt concepts.
1.0	Consolidated Chapters 1-21; terminology normalized; duplicate and out-of-scope requirements removed.

Interpretation Note

This document is a requirements baseline, not authorization to rebuild the application from scratch. Implementation must inspect and preserve the existing Premium Grading System, its data and its proven workflows. Where this SRS and the working system differ, the discrepancy must be documented and resolved through a reviewed migration decision.

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VOLUME I

Executive Summary

PBSMS is an upgrade path from an existing grading application to a unified basic-school platform. It is designed specifically for Nursery, Kindergarten, Primary and Junior High School. The specification treats the student as the central domain entity and connects admissions, enrolment, attendance, assessment, grading, results, official documents, finance, communication and institutional reporting through shared security, audit and workflow services.

The defining management requirement is continuous accountability: weekly, monthly, termly and yearly role-based reports are automatically generated and delivered by email to responsible personnel. These reports show target-versus-actual performance, incomplete responsibilities, risks, recommended actions, owners and deadlines. Acknowledgement, action tracking and escalation close the loop.

The defining implementation rule is preservation. Existing data and working grading functionality must remain intact. Enhancements use safe migrations, versioned policies, server-side permission checks, auditable workflows and phase gates.

Scope at a Glance

Area	Included
Education	Nursery 1-2, Kindergarten 1-2, Basic 1-6, JHS 1-3
Calendar	Three terms; Third Term normally promotion term
Academic Models	Developmental assessment for early years; configurable numerical assessment for Primary and JHS
Governance	Multi-role RBAC, scoped access, configurable approval workflows and immutable audit history
Intelligence	KPIs, scheduled reporting, email delivery, actions and escalation
Historical Control	Versioned grading policies, results and generated documents

CHAPTER 1

Introduction

This chapter defines the authoritative requirements for introduction within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

1.1 Background

Basic schools often maintain admissions, attendance, grading, finance and communication in separate records. The resulting duplication weakens accuracy, delays reporting and makes institutional performance difficult to monitor. PBSMS upgrades the existing Premium Grading System into one integrated platform while preserving working academic records and workflows.

1.2 Problem Statement

The current operating environment is constrained by fragmented data, repeated entry, inconsistent reports, weak cross-department visibility and limited accountability. A unified platform is required to connect school operations without replacing validated functionality or losing historical records.

1.3 Purpose

This specification defines the business, information, functional and architectural requirements for a secure, configurable and auditable management platform for preschool through Junior High School.

1.4 Vision and Mission

PBSMS will be the central operational platform for the school: one trusted source of information, configurable workflows, responsible automation and timely management intelligence.

1.5 Strategic Objectives

- Establish one authoritative source of institutional data.
- Automate repetitive work while retaining human approval for sensitive decisions.
- Preserve complete academic, financial and operational history.
- Provide role-specific dashboards and scheduled performance reports.
- Support safe growth from one campus to multiple campuses.

1.6 Stakeholders

- Proprietor or School Director
- Headmaster and Assistant Headmaster
- Academic Coordinator and Examination Officer
- Admission Officer
- Accountant or Bursar
- Teachers and support staff
- Parents or guardians
- Students
- ICT and system administrators

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 2

Business Context and System Scope

This chapter defines the authoritative requirements for business context and system scope within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

2.1 Business Context

A school manages people, learning, money, documents, communication, facilities and institutional accountability. PBSMS treats these responsibilities as connected business domains supported by a common security, data and reporting foundation.

2.2 Business Drivers

- Operational efficiency
- Academic quality
- Financial accountability
- Institutional governance
- Parent communication
- Data integrity and retention

2.3 In-Scope Education Levels

- Nursery 1 and Nursery 2
- Kindergarten 1 and Kindergarten 2
- Basic 1 through Basic 6
- JHS 1 through JHS 3

2.4 Academic Calendar

The default structure uses First Term, Second Term and Third Term. Third Term is normally the promotion term. Names, dates and progression remain configurable.

2.5 Out of Scope

- Senior High School administration
- University, college or tertiary structures
- Semesters, credit hours, GPA or CGPA
- Lecturers, faculties and degree programmes
- University course registration and graduation clearance

2.6 Ghana Defaults

The default currency is GHS (Ghana cedi) and the default time zone is Africa/Accra. Both are configuration values, not hard-coded assumptions.

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 3

Business Requirements

This chapter defines the authoritative requirements for business requirements within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

3.1 Primary Goal

Provide one integrated, secure platform for the academic, administrative, operational and financial management of a Nursery-to-JHS school.

3.2 Strategic Requirements

- BR-001: Maintain one authoritative source of institutional data.
- BR-002: Reduce manual workload through safe automation.
- BR-003: Provide timely and role-appropriate decision information.
- BR-004: Support transparent, reconcilable finance operations.
- BR-005: Improve communication with staff and guardians.
- BR-006: Enforce accountability through permissions, workflows and audit trails.
- BR-007: Preserve historical records and published versions.
- BR-008: Support configurable policies without source-code changes.
- BR-009: Monitor responsibilities using measurable KPIs.
- BR-010: Scale without architectural redesign.

3.3 Success Measures

Success is measured by preserved legacy data, reduced duplicate entry, complete permission enforcement, faster processing, reconciled finance, timely reporting, traceable approvals and successful role-based email delivery.

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 4

Enterprise Design Principles

This chapter defines the authoritative requirements for enterprise design principles within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

4.1 Single Source of Truth

Each business entity has one authoritative owner. Other modules consume its data through controlled services.

4.2 Modular Independence

Modules own their business responsibility and may evolve without direct manipulation of another module's internal data.

4.3 Configuration Before Customization

Calendars, class levels, subjects, grading, assessments, fees, reports and workflows are configured in the application.

4.4 Security by Design

Authentication, server-side authorization, record scope, auditing and sensitive-data controls apply to every protected operation.

4.5 Automation with Oversight

Routine calculations, reminders, schedules and document generation are automated. Admissions, publication, reversals, promotion and elevated access retain approval gates.

4.6 Historical Preservation

Published academic results, payments, receipts, enrolments and audit records are never silently overwritten or deleted.

4.7 Usability and Accessibility

Interfaces are responsive, keyboard usable, clearly labeled and consistent across roles.

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
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CHAPTER 5

Enterprise Operating Model

This chapter defines the authoritative requirements for enterprise operating model within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

5.1 Institutional Administration

- School configuration
- Identity and access
- Academic calendars
- Policies and system governance

5.2 Student Lifecycle

- Admissions
- Student and guardian records
- Enrolment and transfers
- Promotion and JHS completion

5.3 Academic Operations

- Classes and subjects
- Teacher assignments
- Timetables and attendance
- Assessment, grading, results and report cards

5.4 Financial Management

- Fee structures
- Invoices
- Payments and receipts
- Reversals and financial reports

5.5 Communication and Intelligence

- Notifications and email
- KPI monitoring
- Scheduled reporting
- Action tracking and escalation

5.6 Governance and Compliance

- Audit trails
- Security monitoring
- Backup and recovery
- Document retention and system health

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 6

Enterprise System Architecture

This chapter defines the authoritative requirements for enterprise system architecture within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

6.1 Architectural Objectives

- Modularity and loose coupling
- Scalability and maintainability
- Reliable automation
- Secure integrations
- High availability and recovery readiness

6.2 Logical Layers

- Presentation layer: dashboards, forms and reports.
- Application layer: requests, validation and transactions.
- Business domain layer: institutional rules and workflows.
- Integration layer: internal APIs, events and external connectors.
- Data access layer: repositories, queries, caching and transactions.
- Database layer: master, transactional, historical and audit data.

6.3 Enterprise Domains

- Foundation Services
- Student Lifecycle
- Academic Operations
- Financial Management
- Operational Intelligence
- Governance and Compliance

6.4 Communication Rules

Modules communicate through approved services, APIs or events. Direct cross-module table manipulation is prohibited.

6.5 Event-Driven Processing

Important events such as Admission Approved, Result Published, Student Promoted and Payment Reversed trigger controlled downstream actions without duplicated manual entry.

6.6 Deployment Readiness

The logical design supports cloud, on-premise and hybrid deployment while preserving the existing project stack unless a reviewed migration is justified.

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 7

Master Data Architecture

This chapter defines the authoritative requirements for master data architecture within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

7.1 Purpose

Master data provides consistent reference values shared by operational modules.

7.2 Institutional Master Data

- School and campuses
- Academic years and terms
- Divisions and class levels
- Classes, rooms, calendars and holidays

7.3 Academic Master Data

- Subjects and learning areas
- Assessment types and configurations
- Grading scales
- Promotion rules
- Report templates

7.4 People and Finance Reference Data

- Relationships, nationalities and languages
- Employment types and job titles
- Fee categories and payment methods
- Scholarship and discount types

7.5 Governance

Every master record includes identity, name, status, effective dates, creator, modifier and timestamps. Changes require permission and produce audit events.

7.6 Versioning

Policies used by published academic or financial records remain historically attached even when later configuration changes.

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.

Control	Requirement
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 8

Enterprise Data Architecture

This chapter defines the authoritative requirements for enterprise data architecture within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

8.1 Data Classes

- Master data
- Transactional data
- Reference data
- Generated data
- Audit and security data

8.2 Ownership Model

A business entity has one owner and any number of authorized consumers. Ownership controls mutation; permissions control visibility.

8.3 Integrity

- Entity integrity through immutable primary identifiers.
- Referential integrity through foreign keys.
- Domain integrity through validation and controlled vocabularies.
- Business integrity through workflow and policy checks.

8.4 Lifecycle

Records move through controlled states such as Draft, Submitted, Reviewed, Approved, Published, Locked and Archived. Every transition is authorized and audited.

8.5 Retention

Student profiles, enrolments, attendance, scores, results, report cards, promotions, financial transactions and audit logs are retained according to policy and normally archived rather than deleted.

8.6 Security Classification

- Public
- Internal
- Confidential
- Highly Confidential

8.7 Sensitive Data

Medical details, identification numbers, salaries, banking data, password material and security settings require elevated controls, masking and restricted export.

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 9

Enterprise Data Flow Architecture

This chapter defines the authoritative requirements for enterprise data flow architecture within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

9.1 Admission Flow

Application -> document review -> interview or assessment -> approval -> student profile -> guardian link -> enrolment -> class assignment -> optional invoice -> notification -> audit.

9.2 Academic Flow

Academic year -> term -> class and subject setup -> teacher assignment -> enrolment -> attendance -> assessment -> grading -> result approval -> publication -> report card.

9.3 Promotion Flow

Approved Third-Term results -> annual calculation -> rules -> recommendation -> review -> approval -> next-year enrolment -> prior enrolment closure.

9.4 Finance Flow

Fee structure -> invoice -> payment -> allocation -> receipt -> balance update -> notification -> reconciled report.

9.5 Intelligence Flow

Operational events -> KPI calculations -> scheduled report -> secure email -> acknowledgement -> action -> escalation -> dashboard update.

9.6 Validation and Audit

Every flow validates actor permissions, record scope, dates, duplicate risk and current state before committing an auditable transaction.

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 10

Enterprise Business Rules Framework

This chapter defines the authoritative requirements for enterprise business rules framework within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

10.1 Institutional Rules

- Only one academic year is normally current per school or campus.
- Term dates fall within the academic year.
- Every class belongs to one level and every level to one division.
- Every offered class subject must have a valid configuration before publication.

10.2 Student Rules

- Student and admission identifiers are unique and never reused.
- A student has no more than one active enrolment per academic year unless an explicit exception is configured.
- Transfers preserve academic, attendance, financial, health and discipline history.
- Student records are archived, not permanently deleted.

10.3 Attendance Rules

- Only active enrolments appear in registers.
- One record exists per student, date and configured session.
- Late changes require permission and a reason.
- Absence thresholds generate alerts.

10.4 Assessment and Result Rules

- Teachers enter scores only for active assignments.
- Scores remain within configured bounds.
- Missing scores require a status or reason.
- Submitted and published data is locked until a controlled reopening.
- Publication requires all configured approvals.

10.5 Finance Rules

- Invoices and receipts have unique numbers.
- Payments reference valid obligations and are never deleted.
- Corrections use reversal transactions with approval and reason.
- Reports reconcile to source transactions.

10.6 Reporting Rules

- Reports respect scope and least privilege.
- Schedules include weekly, monthly, termly and yearly cycles.
- Reports identify issues, impact, responsible officer, priority and deadline.
- Failed deliveries retry and escalate.

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 11

Role, Responsibility and Authority Architecture

This chapter defines the authoritative requirements for role, responsibility and authority architecture within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

11.1 Model

PBSMS separates roles (institutional identity), responsibilities (expected work), permissions (allowed actions) and authority (approval power).

11.2 Standard Roles

- Proprietor or School Director
- School Administrator
- Headmaster and Assistant Headmaster
- Academic Coordinator and Examination Officer
- Admission Officer
- Accountant or Bursar
- Class, subject, nursery and kindergarten teachers
- Librarian, Health Officer, Storekeeper and Transport Officer
- Parent or Guardian
- Student
- ICT Officer or System Administrator

11.3 Permission Actions

- View
- Create
- Edit
- Submit
- Review
- Return
- Approve
- Reject
- Publish
- Reopen
- Reverse
- Export
- Print
- Configure
- Archive
- Restore
- View audit history

11.4 Scope

- School
- Campus
- Department
- Division
- Class level
- Class
- Subject
- Assigned students
- Linked children
- Record ownership

11.5 Delegation

Temporary authority includes delegator, recipient, exact functions, start, end and automatic expiry. Delegation is visible and audited.

11.6 Conflict of Interest

Workflows identify when one user holds overlapping roles and prevent required independent review from being silently bypassed.

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 12

Operational Intelligence Framework

This chapter defines the authoritative requirements for operational intelligence framework within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

12.1 Purpose

Operational Intelligence turns activity data into measurable performance, scheduled reports, assigned actions and accountable follow-up.

12.2 Components

- KPI Engine
- Reporting Engine
- Scheduler
- Email Service
- Notification Engine
- Action Tracker
- Escalation Engine
- Executive Dashboards

12.3 KPI Definition

- Code and name
- Responsible role
- Data source and formula
- Target and weight
- Warning and critical thresholds
- Reporting frequency
- Supervisor and status

12.4 Reporting Frequencies

- Daily for immediate operations.
- Weekly for short-cycle responsibilities and exceptions.
- Monthly for trends and target progress.
- Termly for comprehensive academic and operational review.
- Yearly for institutional performance and planning.

12.5 Report Content

- Executive summary
- Target versus actual
- Previous-period comparison
- Completed, incomplete and overdue responsibilities
- Risks and warnings
- Recommended actions

- Responsible personnel
- Priority and deadline
- Secure link to details

12.6 Email Delivery

Reports are role-scoped, sent by schedule, optionally attach safe PDFs, use authenticated links for sensitive detail, record delivery state and retry failures.

12.7 Acknowledgement and Actions

Recipients acknowledge reports, comment, assign owners, set deadlines, attach evidence, complete or reopen actions.

12.8 Escalation

Overdue issues advance through configurable supervision levels and remain visible in subsequent reports until resolved.

12.9 AI Readiness

Future AI may summarize trends and recommend interventions, but official decisions remain with authorized personnel.

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 13

User, Role, Permission and Responsibility Management

This chapter defines the authoritative requirements for user, role, permission and responsibility management within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

13.1 Identity Architecture

Every person has a unique digital identity linked, where appropriate, to a staff, guardian or student profile. Identities are never reused.

13.2 Account Lifecycle

Requested -> verified -> created -> role assigned -> activated -> first-login change -> normal use -> suspended or archived.

13.3 Security Controls

- Secure password hashing
- Failed-login tracking and lockout
- Session timeout and revocation
- Optional MFA
- Login history
- Password reset and first-login change

13.4 Multiple Roles

A user can hold several roles simultaneously. Effective permissions are calculated from active assignments and scopes.

13.5 Role Dashboards

Dashboards combine authorized responsibilities from all active roles without duplicating accounts.

13.6 Administration

- Create, copy and archive roles.
- Assign temporary roles with dates.
- Preview effective permissions.
- Inspect permission history.
- Restrict by campus, division, class or subject.

13.7 Performance Linkage

Responsibilities such as attendance submission, score entry, approval turnaround and reconciliation are measurable and feed role-based reports.

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 14

Admissions Management

This chapter defines the authoritative requirements for admissions management within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

14.1 Purpose

Admissions controls the applicant lifecycle before any permanent student record is created.

14.2 Applicant Data

- Identity and photograph
- Birth, nationality, language and address
- Previous school and class applied for
- Guardian and emergency contacts
- Medical and learning-support information
- Supporting documents
- Interview or entrance-assessment outcomes

14.3 States

- Draft
- Submitted
- Under Review
- Documents Incomplete
- Interview or Assessment Scheduled
- Waitlisted
- Approved
- Rejected
- Admitted
- Cancelled

14.4 Workflow

Application -> documents -> review -> interview or assessment -> decision -> conversion -> admission number -> guardian link -> enrolment -> optional invoice -> accounts and notification.

14.5 Duplicate Prevention

Potential matches use configurable combinations of name, birth date, guardian contact and prior school. Matches are reviewed, not automatically discarded.

14.6 Conversion

Approval performs one transactional conversion so that student, relationships and enrolment are created together or not at all.

14.7 Reports and KPIs

- Applications received and processed
- Incomplete applications
- Processing time
- Conversion rate
- Target versus actual
- Pending interviews and documents

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 15

Student Lifecycle Management

This chapter defines the authoritative requirements for student lifecycle management within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

15.1 Central Record

One permanent student profile connects admissions, guardians, enrolments, attendance, academics, finance, health, discipline, documents, promotion and JHS completion.

15.2 Student Data

- Permanent identifier and admission number
- Identity and contact details
- Guardian relationships
- Current and historical enrolments
- Medical alerts and learning support
- Academic, attendance and financial summaries
- Documents and timeline

15.3 Enrolment

Academic-year enrolment is the source of class membership. Activation exposes the learner to registers, assessments, timetable, finance and authorized teachers.

15.4 Guardian Relationships

A student may have multiple guardians and a guardian multiple students. Relationship-level flags control primary contact, emergencies, pickup, finance and report access.

15.5 Transfers

Class, campus, inbound and outbound transfers use effective dates, reasons, documents and approval while preserving history.

15.6 Health and Discipline

Health information is more restricted than general student data. Discipline supports investigation, response, guardian contact, follow-up, appeal and positive behaviour.

15.7 Promotion and Completion

Third-Term processing creates promotion decisions and new enrolments. JHS 3 transitions to Completed Basic Education without deleting the record.

15.8 Student Timeline

Every significant event is recorded chronologically with timestamp, actor, module, description and related evidence.

15.9 Parent Portal

Guardians see only linked learners and only published or authorized academic, attendance, financial and communication records.

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 16

Academic Management

This chapter defines the authoritative requirements for academic management within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

16.1 Academic Hierarchy

School -> campus -> academic year -> term -> division -> class level -> class or section -> class subjects -> teacher assignments.

16.2 Default Progression

Nursery 1 -> Nursery 2 -> Kindergarten 1 -> Kindergarten 2 -> Basic 1 through Basic 6 -> JHS 1 through JHS 3 -> Completed Basic Education.

16.3 Academic Year and Terms

Years and terms use planned, active/open, closed and archived states. Score-entry, review, approval and publication windows are separately configurable.

16.4 Classes and Sections

- Code and name
- Level and academic year
- Room and capacity
- Class teacher and assistant
- Student count and assignment history

16.5 Subjects and Learning Areas

Subjects vary by division and class level. Nursery and Kindergarten may use developmental learning areas; Primary and JHS use configurable subject structures.

16.6 Teacher Assignments

Teachers may teach multiple classes and subjects. Assignments include dates, scope and status; duplicates and timetable conflicts are detected.

16.7 Timetable

A published timetable prevents teacher, class and room conflicts and provides class, teacher and room views.

16.8 Policies

- Assessment policy
- Attendance policy
- Promotion policy
- Examination policy
- Configurable ranking policy

16.9 Reports

- Teacher workload
- Unassigned subjects
- Timetable compliance
- Academic progress
- Curriculum coverage
- Promotion readiness

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 17

Attendance Management

This chapter defines the authoritative requirements for attendance management within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

17.1 Scope

Attendance covers students, teachers and non-teaching staff while contributing to communication, promotion and performance monitoring.

17.2 Policies

- Instructional calendar
- Submission windows
- Late and early thresholds
- Excused absence rules
- Minimum attendance targets

17.3 Student Register

Registers are generated from active enrolments for valid instructional days and configured sessions.

17.4 Statuses

- Present
- Absent
- Late
- Excused
- Sick
- Suspended
- Official Activity

17.5 Corrections

After the submission window, changes require permission, reason and approval while retaining original and corrected values.

17.6 Alerts

Repeated absence, low attendance and missing teacher submissions generate notifications, KPI updates and escalation when unresolved.

17.7 Staff Attendance

Check-in, check-out, leave, official duty, lateness and corrections support daily through yearly summaries.

17.8 Reports

- Daily registers
- Weekly trends

- Monthly class and staff summaries
- Term promotion eligibility
- Yearly institutional attendance analysis

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 18

Assessment Management

This chapter defines the authoritative requirements for assessment management within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

18.1 Separation of Concerns

Assessment captures observations and scores; it does not define official grades or published results.

18.2 Nursery and Kindergarten

- Developmental domains and indicators
- Teacher observation
- Participation and behaviour
- Communication, motor, literacy and numeracy readiness
- Configurable ratings such as Emerging, Developing, Achieving and Exceeding Expectations

18.3 Primary and JHS

- Exercises, homework, projects, quizzes and tests
- Practical and oral work
- Mid-term and end-of-term examinations
- Configurable component weights

18.4 Configuration

Assessment structures may vary by division, level, subject, academic year and term. Weights must satisfy the configured total.

18.5 Score Entry

Teachers see assigned classes and active enrolments, save drafts, bulk enter or import authorized files, set missing-score statuses and submit.

18.6 Validation

- No score below zero or above maximum.
- Only valid assignments and enrolments.
- No duplicate student-component record.
- Closed windows reject entry.
- Missing scores require a status or reason.

18.7 Moderation

Optional review records moderator, findings, requested corrections, comments and decision before data is released to grading.

18.8 KPIs and Reports

- Completion rate

- Submission timeliness
- Missing assessments
- Correction frequency
- Moderation turnaround

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 19

Enterprise Grading Engine

This chapter defines the authoritative requirements for enterprise grading engine within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

19.1 Purpose

The grading engine interprets approved assessment data using a versioned policy. It never collects scores directly.

19.2 Models

- Developmental grading for Nursery and Kindergarten.
- Numerical grading for Primary and JHS.
- Future competency-based models without changing historical outcomes.

19.3 Scale

Each scale item has minimum, maximum, grade, optional point, remark and pass/fail status. Ranges cannot overlap or leave unintended gaps.

19.4 Processing

Approved scores -> totals -> percentage -> policy lookup -> grade -> remark -> aggregate calculations -> result candidate.

19.5 Aggregates

- Subject total and grade
- Term average
- Annual average
- Subjects passed and requiring support
- Promotion recommendation

19.6 Ranking

Ranking is disabled by default where unsuitable and configurable for Primary or JHS. Supported modes include competition, dense and none, with explicit tie rules.

19.7 Exceptions

Undefined scales, incomplete components and invalid weights stop processing and create actionable errors.

19.8 Historical Accuracy

Every published outcome retains the exact grading-policy version used.

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 20

Results Management Engine

This chapter defines the authoritative requirements for results management engine within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

20.1 Purpose

Results Management transforms graded data into the school's official academic outcome through validation, approval, publication and version control.

20.2 Compilation

The engine verifies enrolment, required subjects, assessment approval, grading policy and completeness before assembling subject and overall results.

20.3 Workflow

Subject Teacher -> Class Teacher -> optional Academic Coordinator or Examination Officer -> Headmaster -> authorized publication.

20.4 States

- Draft
- Submitted
- Returned
- Corrected
- Reviewed
- Approved
- Published
- Locked
- Archived

20.5 Publication Rules

- All configured approvals complete.
- Required subjects resolved or formally exempted.
- Term and publication windows valid.
- Applicable report template available.
- No unresolved validation errors.

20.6 Corrections and Versioning

Published results are never overwritten. Reopening requires reason and authority; recalculation creates a new version while preserving the previous publication.

20.7 Access

Teachers see assigned records, managers see their authorized scope, guardians see linked learners, and students see only their own published results.

20.8 Analytics

- Student trends
- Class average and pass rate
- Subject performance
- Division comparisons
- Promotion readiness

20.9 Audit Events

- Compiled
- Submitted
- Returned
- Approved
- Published
- Viewed
- Exported
- Reopened
- Corrected
- Archived

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

CHAPTER 21

Enterprise Academic Document Engine

This chapter defines the authoritative requirements for enterprise academic document engine within PBSMS. Requirements apply only to the Nursery-to-JHS product scope and must be implemented consistently with the preservation, security and audit principles established in this volume.

21.1 Purpose

EADE is the shared service for composing, branding, generating, verifying, distributing, versioning and archiving official institutional documents.

21.2 Document Categories

- Nursery and Kindergarten development reports
- Primary and JHS report cards
- Admission and transfer letters
- Student identity cards and enrolment confirmations
- Invoices, receipts and statements
- Timetables, registers and calendars
- KPI, audit and executive reports

21.3 Templates

Templates define header, footer, logo, colours, typography, margins, signatures, verification area, watermark and pagination. Authorized configuration replaces hard-coded layouts.

21.4 Report Cards

Nursery and Kindergarten emphasize developmental domains, observations, attendance, strengths and support needs without compulsory numerical ranking. Primary and JHS include configured components, totals, grades, remarks, attendance and promotion decisions.

21.5 Generation Workflow

Approved source -> template and policy snapshot -> branding -> signatures -> verification token -> PDF -> archive -> secure delivery.

21.6 Verification

QR codes or tokens confirm document identity, type, issue date and authenticity without exposing confidential details to unauthenticated viewers.

21.7 Signatures

Manual, stored or electronic approval signatures are versioned and applied according to document policy.

21.8 Batch Processing

Authorized users may generate class or school batches with progress, retry and failure reporting.

21.9 Archive and Versioning

Generated documents are immutable versions linked to their source snapshot. Reissuance produces a new version, never an overwrite.

21.10 Distribution

- Parent and student portals
- Secure authenticated download
- Email with safe attachment or secure link
- Administrative print queue

21.11 Audit Events

- Template created or changed
- Document generated
- Viewed
- Downloaded
- Printed
- Verified
- Reissued
- Archived

Chapter Controls

Control	Requirement
Permissions	Every protected action requires server-side authorization and record-scope validation.
Audit	Material state changes record actor, time, old and new values, reason and outcome.
History	Official records use versioning or reversal; they are not silently overwritten.
Reporting	Applicable operations feed role-based dashboards, scheduled reports and follow-up actions.
Testing	Business rules, permissions, state transitions, calculations and regressions require automated and manual verification.

APPENDIX A

Consolidated Role and Reporting Matrix

Role	Primary monitoring focus	Scheduled reporting
Proprietor / Director	Institutional performance, finance, risks and unresolved actions	Monthly, termly, yearly; urgent exceptions
Headmaster	Academics, staff, attendance, approvals and school risks	Weekly, monthly, termly, yearly
Academic Coordinator	Assignments, assessment completion, missing scores and interventions	Weekly, monthly and termly
Admission Officer	Applications, missing documents, processing time and conversion	Weekly and monthly
Accountant / Bursar	Collections, balances, reversals, reconciliation and targets	Daily digest, weekly, monthly, termly, yearly
Class Teacher	Attendance, at-risk learners, remarks and parent follow-up	Weekly and termly
Subject Teacher	Score completion, missing scores, returns and deadlines	Weekly and termly
Librarian / Health / Support	Role-specific operational exceptions and actions	Weekly or monthly as configured

APPENDIX B

Core State Models

Entity	Canonical states
Admission	Draft -> Submitted -> Review -> Incomplete/Waitlisted -> Approved/Rejected -> Admitted
Assessment	Planned -> Open -> Draft -> Submitted -> Moderated -> Approved -> Locked
Result	Draft -> Submitted -> Returned -> Corrected -> Reviewed -> Approved -> Published -> Locked -> Archived
Invoice	Draft -> Issued -> Partially Paid -> Paid/Overdue -> Cancelled/Written Off by approval
Action	Open -> Assigned -> In Progress -> Completed -> Closed; may become Overdue or Escalated
Document	Generated -> Verified -> Published -> Superseded/Archived

State-Control Rule

Transitions are explicit business events. Each transition validates permissions, the current state and required evidence; it records a timestamp and actor; and it triggers relevant notifications, reporting updates and audit events.

APPENDIX C

Phased Upgrade Controls

The SRS describes the target product. Delivery must proceed through controlled phases so the existing Premium Grading System remains usable and recoverable.

Phase	Primary outcome
1 - Inspection and Protection	Repository, database, build, tests, risks and backups verified.
2 - Foundation and Security	Authentication, roles, scopes, audit and notification foundations.
3 - School and Academic Setup	School, year, term, divisions, levels, classes, subjects and assignments.
4 - Student Lifecycle	Admissions, students, guardians, enrolments, transfers and documents.
5 - Academic Operations	Attendance, timetable, assessments, score entry, grading and remarks.
6 - Results and Completion	Approvals, publication, documents, promotion and JHS completion.
7 - Finance	Fees, invoices, payments, reversals and financial reports.
8 - Intelligence	KPIs, scheduled email reports, acknowledgement, actions and escalation.
9 - Supporting Operations	Library, transport, health, discipline, inventory and communication.
10 - Technical Completion	Backups, system health, hardening, testing, deployment and handover.

Phase Gate

A phase does not advance until applicable migrations, regression tests, permissions, audit behavior, scheduled jobs, build, type checking, linting, desktop and mobile layouts, backup reference and rollback instructions have been verified. Known issues must be reported honestly.

APPENDIX D

Glossary and Terminology

Term	Meaning
PBSMS	Premium Basic School Management System.
Active enrolment	The authoritative academic-year class membership for a student.
Developmental assessment	Observation- and competency-based early-years assessment.
Published result	An approved, versioned official academic outcome.
Record scope	The school, campus, division, class, subject or relationship boundary limiting access.
KPI	A measurable indicator comparing responsibility performance with a target.
Operational Intelligence	The integrated KPI, reporting, scheduling, email, action and escalation subsystem.
Reversal	A controlled compensating financial transaction; not deletion.
EADE	Enterprise Academic Document Engine.
JHS completion	Completion of Basic Education after JHS 3; not a tertiary graduation model.

End of Volume I

This consolidated volume ends at Chapter 21. Later volumes may define remaining functional domains, detailed database and API contracts, UI standards, security and deployment architecture, traceability, tests and the phased Codex implementation blueprint. Those materials are not represented as completed in this volume.

PREMIUM BASIC SCHOOL MANAGEMENT SYSTEM

Enterprise Software Requirements Specification

VOLUME II

Functional Modules and Processing Engines
Chapters 22-31

Nursery - Kindergarten - Primary - Junior High School
Version 1.0 | July 2026

Document Control

Item	Value
Document	PBSMS Enterprise SRS - Volume II
Scope	Functional Modules and Processing Engines
Chapters	22 through 31
Education levels	Nursery through JHS
Default currency	GHS (Ghana cedi)
Version	1.0
Status	Consolidated specification
Date	July 2026

Purpose

This volume continues Volume I and defines the processing engines and operational modules required to turn the enterprise foundation into controlled day-to-day school operations. It is an implementation-facing requirements document, but it remains technology-neutral.

Interpretation

The words shall and must identify mandatory requirements. May identifies configurable or optional behavior. Future capability is not part of initial acceptance unless separately approved.

Scope Boundary

PBSMS is designed for Nursery, Kindergarten, Primary and Junior High School. Senior High School, tertiary semesters, credit hours, GPA and university structures are outside this volume.

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CHAPTER 22

22. Promotion and JHS Completion Engine

22.1 Purpose and Scope

The Promotion and JHS Completion Engine converts approved annual academic evidence into controlled progression decisions. It supports promotion, repetition, conditional promotion, pending decisions, transfer outcomes and completion of Basic Education after JHS 3 without overwriting historical enrolments.

22.2 Objectives

- Apply configurable promotion policies by division and class level.
- Separate system recommendations from authorized decisions.
- Create the next academic-year enrolment only after approval.
- Preserve performance, attendance, conduct and decision evidence.
- Support JHS completion and alumni-ready records.

22.3 Promotion Policy

- Applicable academic year, division and class level
- Minimum annual average and subjects passed
- Required performance in named core subjects
- Minimum attendance percentage
- Conduct or intervention conditions where permitted
- Automatic recommendation flag and mandatory approval route
- Effective date, version and status

22.4 Evidence and Calculation

The engine uses only approved, versioned results and the attendance record applicable to the same academic period. Annual weighting by term is configurable. Missing terms, withheld results or unresolved corrections stop automatic recommendation and create an exception.

22.5 Decision Outcomes

Outcome	Meaning	System effect
Promoted	All applicable conditions satisfied	Prepare next-level enrolment
Conditionally Promoted	Authorized exception with conditions	Prepare enrolment and intervention
Repeated	Remain at current level	Prepare same-level enrolment
Pending	Evidence or approval incomplete	No new enrolment
Transferred	Leaves progression route	Follow transfer workflow
Completed Basic Education	Approved JHS 3 completion	Create completion record

22.6 Workflow

Results approved -> annual performance calculated -> rule set applied -> recommendation produced -> class teacher review -> academic review -> headmaster approval -> next enrolment created -> prior enrolment closed -> notifications and documents generated.

22.7 Review and Approval Rules

- The reviewer sees the evidence and rule outcome, not only a recommendation.
- Overrides require a reason and appropriate authority.
- A user cannot provide independent approval where a conflict-of-interest rule applies.
- Bulk approval must still validate every learner independently.
- Rejected or returned cases re-enter the queue with comments.

22.8 JHS Completion

JHS 3 completion records the completion date, final class, academic year, eligibility evidence, clearance state, approvers and issued documents. Academic history remains available; completion never deletes the student profile.

22.9 Exceptions and Recovery

- Incomplete or disputed results
- Attendance under review
- Student transfer during the year
- Repeat decision after a prior conditional promotion
- Promotion executed against the wrong class
- Failure while creating new enrolments

22.10 Dashboards, Reports and KPIs

- Promotion readiness by class
- Pending and exceptional decisions
- Approved promotion and repetition counts
- JHS completion readiness
- Decision turnaround time
- Percentage of cases overridden
- New-enrolment creation failures

22.11 Security, Audit and Testing

Permissions distinguish calculate, recommend, review, approve, execute, reverse and export. Audit events capture policy version, evidence, old and new decision, reason, user and timestamp. Tests cover boundary scores, missing evidence, duplicate execution, rollback and cross-campus isolation.

Chapter Control Summary

This chapter is complete when its data ownership, permission boundaries, workflow states, notifications, audit events, reports and acceptance tests are traceable to the technical implementation.

CHAPTER 23

23. Academic Analytics Engine

23.1 Purpose

The Academic Analytics Engine transforms authorized student, attendance, assessment, grading and result data into explainable insights for teachers and leaders. It supports intervention and planning; it does not replace professional judgment.

23.2 Principles

- Use approved and contextually comparable data.
- Show the source period, cohort and calculation method.
- Distinguish correlation from causation.
- Avoid unfair ranking across incompatible assessment structures.
- Apply least-privilege access and suppress small sensitive groups.
- Make every automated flag explainable.

23.3 Levels of Analysis

Level	Examples	Primary users
Student	growth, strengths, gaps, attendance relationship	teachers, guardians
Class	average, distribution, missing work, trend	class teacher, coordinator
Subject	pass rate, difficult outcomes, improvement	subject teacher, HOD
Division	cohort patterns and promotion readiness	academic leadership
School or campus	longitudinal trends and comparisons	executives

23.4 Early Years Analytics

Nursery and Kindergarten analytics prioritize developmental progress, observation completeness and support needs. Numerical league tables and academic ranking are disabled by default.

23.5 Primary and JHS Analytics

- Subject and overall averages
- Grade and pass distributions
- Growth from prior comparable periods
- Core-subject performance
- Missing or withheld outcomes
- Promotion readiness
- Performance bands and intervention lists

23.6 Early Warning and Intervention

A rule may flag declining achievement, repeated absence, missing assessment components or continuing failure. Each alert records the evidence, threshold, priority, recommended action, responsible role, due date and review status.

23.7 Teacher and Process Analytics

Teacher analytics focus on operational responsibilities such as score-entry completion, submission timeliness, correction frequency and assigned-class progress. Student results are contextual evidence and must not be used alone as a teacher performance judgment.

23.8 Data Quality Controls

- Completeness checks before aggregation
- Version alignment for published results
- Cohort-size and assessment-structure warnings
- Exclusion of invalid or cancelled records
- Visible refresh timestamp and snapshot period

23.9 Dashboards and Reports

- Student growth profile
- Class performance and attendance relationship
- Subject trend analysis
- At-risk learner queue
- Promotion readiness
- Weekly exception digest
- Termly academic review
- Yearly institutional trend report

23.10 Security and Audit

Parents see only linked children; teachers see assigned learners; leadership access follows campus and division scope. Exports and access to individual profiles are logged. Sensitive intervention notes are restricted.

23.11 Acceptance Criteria

- Recalculation produces repeatable values.
- Every flag links to supporting evidence.
- Unauthorized drill-downs are denied.
- Published snapshots remain stable.
- Small-group privacy rules are enforced.
- Charts and tables reconcile with source results.

Chapter Control Summary

This chapter is complete when its data ownership, permission boundaries, workflow states, notifications, audit events, reports and acceptance tests are traceable to the technical implementation.

CHAPTER 24

24. Enterprise Workflow Engine

24.1 Purpose

The Enterprise Workflow Engine provides one configurable approval and responsibility framework for admissions, results, promotions, financial assistance, payment reversals, leave, discipline, documents and other controlled processes.

24.2 Workflow Definition

- Name, code and owning module
- Trigger event and eligible record types
- Sequential, parallel and conditional stages
- Role, user or scoped assignment rules
- Allowed decisions and required comments
- Deadlines, reminders and escalation
- Completion, cancellation and failure actions
- Version, effective date and status

24.3 Runtime Model

A workflow definition creates an immutable workflow instance for a specific business record. Each stage creates work items. The runtime records assignment, decisions, comments, delegation, elapsed time and completion state.

24.4 Stage Types

Type	Behavior	Example
Sequential	Next stage waits for current completion	Teacher then coordinator
Parallel	Several reviews occur together	Academic and finance clearance
Conditional	Route depends on record data	High-value waiver route
Automatic	System performs validated action	Generate notification
Final	Commits approved business outcome	Publish result

24.5 Actions

- Submit
- Claim
- Approve
- Reject
- Return for correction
- Hold
- Delegate
- Escalate
- Cancel
- Withdraw
- Complete

24.6 Assignment and Delegation

Assignments may target a role, named user, supervisor, campus, division, class or department. Delegation has explicit scope and dates, cannot expand authority, expires automatically and remains auditable.

24.7 Deadlines and Escalation

The engine calculates due dates using configured calendars, sends reminders, marks overdue work and escalates through approved supervision paths. Escalation changes visibility or assignment but does not silently approve a decision.

24.8 Conflict Controls

- Prevent an initiator from independently approving the same request where segregation is required.
- Detect when a class teacher and subject teacher are the same person.
- Require substitute approvers for absent or inactive users.
- Reject routing to users outside the record's campus or scope.

24.9 Versioning

New workflow definitions affect future instances only. Existing instances continue on the version with which they began unless an authorized migration is explicitly executed and logged.

24.10 Administration Interface

- Visual ordered stages
- Condition builder
- Role and scope selector
- Deadline and escalation setup
- Notification template mapping
- Test simulation
- Draft, publish and archive states
- Instance monitor and stuck-work recovery

24.11 Dashboards and Reports

- My pending work
- Team queues
- Overdue stages
- Average turnaround
- Return and rejection rate
- Delegations
- Failed completion actions
- Workflow audit report

24.12 Security, Audit and Testing

Only authorized administrators publish definitions. Every state transition and rule evaluation is recorded. Tests cover branching, parallel completion, expiry, delegation, inactive approvers, retries, idempotent completion and permission boundaries.

Chapter Control Summary

This chapter is complete when its data ownership, permission boundaries, workflow states, notifications, audit events, reports and acceptance tests are traceable to the technical implementation.

CHAPTER 25

25. Enterprise Financial Management

25.1 Purpose

Enterprise Financial Management controls student billing, payments, assistance, adjustments, reconciliation and financial intelligence in Ghana cedis by default. It emphasizes transaction integrity, separation of duties and traceable balances.

25.2 Scope

- Finance configuration and numbering
- Fee structures and billing
- Invoices and invoice items
- Payments, allocations and receipts
- Scholarships, discounts, waivers and sponsors
- Credits, adjustments, reversals and write-offs
- Reconciliation, statements and reporting

25.3 Core Principles

- Posted transactions are never silently edited or deleted.
- Every balance is derived from valid transactions.
- Receipt and invoice numbers are unique.
- Sensitive actions require reasons and, where configured, approval.
- Financial reports reconcile to source records.
- Access is restricted by school, campus and role.

25.4 Financial Configuration

Configuration	Examples
Currency	GHS, symbol, decimal precision
Numbering	invoice, receipt, credit and adjustment sequences
Periods	academic year, term and accounting close status
Methods	cash, mobile money, bank transfer, card, cheque
Controls	approval thresholds and cashier limits
Communication	invoice, receipt and overdue templates

25.5 Financial Record Model

Student obligations are represented by invoices and items. Payments remain independent receipts of value and are connected to obligations through allocations. Assistance and adjustments are explicit transactions, enabling a complete explanation of every balance.

25.6 Segregation of Duties

The platform distinguishes configuration, invoice generation, payment capture, verification, reversal approval, assistance approval, reconciliation, period close and report export. Schools may enforce maker-checker controls.

25.7 Operational Flow

Fee structure -> eligible students -> billing preview -> invoice issue -> payment capture -> verification -> allocation -> receipt -> statement update -> reconciliation -> scheduled reporting.

25.8 Dashboards

- Today's collections
- Unverified transactions
- Outstanding and overdue balances
- Collection target versus actual
- Unallocated credits
- Pending reversals and assistance
- Reconciliation differences
- Failed email deliveries

25.9 Security and Privacy

Payment references, sponsor details and financial exports are confidential. Parent access is restricted to linked children. Cashier views expose only the information needed to receive and confirm payment.

25.10 Audit and Recovery

Every posting records the actor, source, time, amount, currency, related student, method and reference. Multi-record operations use database transactions and idempotency protections.

25.11 Acceptance Criteria

- No duplicate invoice or receipt numbers.
- Balances recalculate correctly after each transaction.
- Reversals preserve original records.
- Reports reconcile by period and method.
- Unauthorized cross-campus access is rejected.
- Retrying a financial request does not double-post.

Chapter Control Summary

This chapter is complete when its data ownership, permission boundaries, workflow states, notifications, audit events, reports and acceptance tests are traceable to the technical implementation.

CHAPTER 26

26. Fee Structure and Student Billing Management

26.1 Purpose

This chapter defines how authorized fee rules become accurate student obligations. Structures remain configurable by academic year, term, campus, division, class level and student category.

26.2 Fee Categories

- Tuition
- Admission
- Examination
- Books and learning materials
- Uniform
- Feeding
- Transport
- PTA levy
- Activities
- Custom approved categories

26.3 Fee Structure Fields

- Academic year and term
- Campus, division and class-level eligibility
- Student category
- Fee category and description
- Amount and due date
- Mandatory or optional status
- Instalment permission
- Proration and penalty rules
- Effective dates and status

26.4 Eligibility Resolution

The billing engine selects active enrolments that match all configured dimensions. A preview shows included students, exclusions, amounts, existing invoices and validation warnings before issue.

26.5 Billing Workflow

Create structure -> validate rules -> approve or activate -> select eligible enrolments -> preview -> resolve duplicates -> generate invoice and items -> issue -> notify guardian -> audit.

26.6 Duplicate Prevention

A uniqueness rule links the student, enrolment, fee structure and billing occurrence. Re-running a completed job returns the existing outcome rather than creating a second obligation.

26.7 Instalments and Due Dates

An instalment plan defines sequence, amount or percentage and due date. The sum must equal the charge. Overdue status is calculated from unpaid due amounts, not merely the invoice creation date.

26.8 Proration and Late Admission

Schools may configure full, daily, monthly or percentage proration for late entry or transfer. The applied rule and calculation details are stored with the invoice item.

26.9 Penalties

Penalty rules specify trigger, grace period, fixed or percentage amount, cap and frequency. Penalties are separate invoice items and cannot compound unless explicitly configured and legally permitted.

26.10 Manual Billing and Corrections

Authorized users may create exceptional charges with a reason and evidence. Issued invoices are corrected through credit, cancellation or adjustment workflows; they are not overwritten.

26.11 Dashboards and Reports

- Billing run status
- Unbilled eligible students
- Duplicate warnings
- Invoices issued by class
- Upcoming due amounts
- Overdue balances
- Manual billing report
- Structure utilization

26.12 Security, Audit and Testing

Separate configure, approve, generate, issue, cancel and export permissions. Tests cover eligibility boundaries, transfers, proration, instalments, reruns, inactive enrolments and concurrent billing.

Chapter Control Summary

This chapter is complete when its data ownership, permission boundaries, workflow states, notifications, audit events, reports and acceptance tests are traceable to the technical implementation.

CHAPTER 27

27. Invoice, Payment Allocation and Receipt Management

27.1 Invoice Lifecycle

Draft -> Issued -> Partially Paid -> Paid, with Overdue, Cancelled or Written Off states controlled by transactions and approval. An invoice includes student, enrolment, period, items, gross amount, assistance, adjustments, amount paid and balance.

27.2 Payment Capture

- Student or payer
- Amount and currency
- Payment date
- Method
- Provider or bank reference
- Receiving officer or integration
- Evidence or notes
- Verification status

27.3 Payment Statuses

Status	Meaning
Pending	Captured but not ready to allocate
Verified	Reference or cash confirmed
Posted	Available for allocation and reporting
Partially Allocated	Some value remains as credit
Fully Allocated	Entire amount applied
Reversed	Offset by approved reversal

27.4 Allocation Rules

Payments may apply to one or several invoices. Allocation order may be explicit, oldest due first or configured priority. The sum of allocations cannot exceed the posted payment or the receivable amount.

27.5 Partial and Excess Payments

Partial allocations update invoice status and preserve the outstanding amount. Excess value becomes an identifiable unallocated credit and is never hidden in the invoice balance.

27.6 Receipt Generation

A receipt is created only for a valid posted payment. It includes a unique number, payer, student, amount, method, reference, allocation summary, cashier, issue time and verification code.

27.7 Reversal Workflow

Request reversal -> state reason and evidence -> review segregation and period status -> approve -> create linked reversal transaction -> reverse allocations -> recalculate invoices -> issue notification -> audit.

27.8 Student Statement

The statement presents opening balance, invoices, assistance, adjustments, payments, allocations, reversals, credits and closing balance in chronological order. Totals must reconcile to the ledger.

27.9 Electronic Payment Integration

Provider callbacks require signature verification, idempotency keys, amount and reference checks, timestamp validation and safe retry behavior. Unmatched transactions enter an exception queue.

27.10 Dashboards and Reports

- Collections by method and cashier
- Pending verification
- Unallocated credits
- Partially paid invoices
- Reversed payments
- Receipt reprints
- Provider exceptions
- Daily payment register

27.11 Security, Audit and Testing

Permissions distinguish capture, verify, allocate, print, reverse and approve. Tests cover duplicate callbacks, partial allocations, overpayment, reversal, closed periods, race conditions and statement reconciliation.

Chapter Control Summary

This chapter is complete when its data ownership, permission boundaries, workflow states, notifications, audit events, reports and acceptance tests are traceable to the technical implementation.

CHAPTER 28

28. Financial Assistance and Adjustments

28.1 Purpose

Financial assistance and adjustments manage scholarships, discounts, waivers, sponsorships, credits, corrections and exceptional relief without obscuring the original charge.

28.2 Assistance Types

Type	Basis	Typical use
Scholarship	percentage or fixed	merit or need support
Discount	percentage or fixed	staff child or sibling policy
Waiver	specific obligation	authorized relief
Sponsorship	third-party commitment	organization or donor support
Credit	stored value	overpayment or approved correction

28.3 Application and Approval

Request -> eligibility and document review -> financial-impact preview -> configured workflow -> approval or rejection -> application to eligible invoice items -> notification -> audit.

28.4 Rules and Stacking

Each rule defines eligible fees, period, cap, sponsor, priority and whether it may combine with other assistance. The calculation engine applies rules in a deterministic configured order and prevents total relief from exceeding the eligible charge.

28.5 Renewal and Expiry

Assistance may be one-time, termly or yearly. Renewal requires current eligibility and, when configured, reapproval. Expiry stops future use but preserves prior applications.

28.6 Sponsors

Sponsor records include identity, contact, commitment limit, eligible students, period, invoicing arrangement, amounts applied and outstanding sponsor obligation. Sponsor access is separately restricted.

28.7 Adjustments

Adjustments include approved debit, credit and correction entries. Each states the original record, amount, reason, evidence, approvers and posting period.

28.8 Write-Offs

A write-off recognizes an authorized uncollectible balance. It never pretends payment occurred. Threshold-based approval, documented reason, period control and a complete audit trail are mandatory.

28.9 Reports and KPIs

- Assistance by type and class
- Sponsor commitment and utilization
- Expiring assistance

- Adjustment and write-off register
- Approval turnaround
- Assistance impact on collections
- Cases exceeding configured thresholds

28.10 Security, Audit and Testing

Sensitive need-based data has restricted visibility. Tests cover stacking order, caps, expiry, sponsor limits, partial eligibility, invoice cancellation and attempts to exceed original charges.

Chapter Control Summary

This chapter is complete when its data ownership, permission boundaries, workflow states, notifications, audit events, reports and acceptance tests are traceable to the technical implementation.

CHAPTER 29

29. Financial Reconciliation, Reporting and Analytics

29.1 Purpose

Reconciliation proves that system transactions agree with cash, bank, mobile money and payment-provider evidence. Reporting then presents reliable operational and management information.

29.2 Reconciliation Types

- Cashier or cash-drawer reconciliation
- Bank deposit reconciliation
- Mobile money settlement reconciliation
- Card or gateway reconciliation
- Invoice-to-ledger reconciliation
- Period opening and closing balance reconciliation

29.3 Matching

Automatic matching uses amount, method, reference, date and provider identifiers. Exact matches are confirmed; tolerances and one-to-many cases follow configured rules; unmatched records enter a controlled exception queue.

29.4 Cash Reconciliation

Opening float + cash received - refunds or approved reversals - deposits = expected cash. The responsible officer records counted cash and the system calculates any variance.

29.5 Variance Workflow

Variance detected -> explanation required -> evidence attached -> supervisor review -> correction or acceptance -> action assigned where necessary -> close and audit.

29.6 Period Close

Before closing a day, month, term or year, the system checks unverified payments, open cash sessions, unresolved variances, pending reversals and failed settlements. Override requires elevated authority and reason.

29.7 Standard Reports

Frequency	Core reports
Daily	collections, cashier, method, exceptions, cash position
Weekly	collection trend, overdue movements, open variances
Monthly	billing and collections, aged receivables, assistance, reversals
Termly	fee performance by class, targets, sponsor position
Yearly	institutional financial review and multi-period trend

29.8 Analytics and KPIs

- Collection rate
- On-time payment rate

- Outstanding and overdue value
- Average days to payment
- Assistance utilization
- Reversal rate
- Unallocated-credit value
- Reconciliation completion and variance rate

29.9 Role-Based Delivery

Accountants receive transaction and reconciliation detail; headmasters receive controlled summaries; proprietors receive institutional trends; parents receive only linked-student statements. Scheduled emails use secure links for sensitive details.

29.10 Action and Escalation

Reports identify the issue, amount, owner, priority and deadline. Unresolved variances, overdue reconciliations and failed settlements escalate and remain in subsequent reports.

29.11 Security, Audit and Testing

Financial report snapshots store criteria and generation time. Exports are logged. Tests reconcile every report to source data, verify period locks and confirm that scope restrictions survive filtering and export.

Chapter Control Summary

This chapter is complete when its data ownership, permission boundaries, workflow states, notifications, audit events, reports and acceptance tests are traceable to the technical implementation.

CHAPTER 30

30. Communication and Notification Centre

30.1 Purpose

The Communication and Notification Centre provides a common service for announcements, transactional notifications, scheduled reports, direct messages and emergency alerts across in-app and email channels.

30.2 Communication Types

- Institutional announcements
- Event-triggered notifications
- Scheduled performance and finance reports
- Approval and action reminders
- Guardian communications
- Emergency alerts
- Authorized direct messages

30.3 Recipient Resolution

Recipients may be selected by user, role, school, campus, department, division, class, subject assignment or guardian-student relationship. Resolution occurs at send time and stores the exact recipient list.

30.4 Channels

Channel	Use	Control
In-app	routine and actionable events	read and acknowledgement state
Email	reports, receipts and formal notices	delivery and retry tracking
SMS interface	short urgent notices where enabled	consent and cost controls
Future push	mobile application notifications	device registration

30.5 Templates

Templates contain subject, body, variables, channel, language, sensitivity level and version. A preview validates missing variables and displays the resolved content before activation.

30.6 Scheduling and Digests

Messages may be immediate, scheduled or periodic. Related low-priority notifications can be combined into role-based digests to reduce fatigue. Quiet hours and expiry rules are configurable.

30.7 Email Reporting

Weekly, monthly, termly and yearly reports identify performance, incomplete responsibilities, risks, recommended actions, owners and deadlines. Highly sensitive detail stays behind an authenticated secure link.

30.8 Delivery and Retry

Queued -> processing -> sent -> delivered where supported -> opened where appropriate -> acknowledged. Temporary failures retry with backoff; permanent failure and repeated failure are reported to administrators.

30.9 Acknowledgement and Actions

Important messages can require acknowledgement, comment, assignment, due date, evidence and closure. The action history remains linked to the original communication.

30.10 Privacy and Consent

The service respects communication preferences, guardian relationships, lawful consent and restricted data categories. Medical, security and detailed financial data are not exposed in ordinary email.

30.11 Dashboards and Reports

- Queue health
- Delivery success and failure
- Pending acknowledgements
- Expired notices
- Template usage
- Report delivery by role
- Unsubscribed or invalid addresses
- Escalated communications

30.12 Security, Audit and Testing

Permissions distinguish compose, approve, schedule, send, view delivery logs and resend. Tests cover scope resolution, duplicate prevention, retries, invalid variables, expired links and unauthorized recipient access.

Chapter Control Summary

This chapter is complete when its data ownership, permission boundaries, workflow states, notifications, audit events, reports and acceptance tests are traceable to the technical implementation.

CHAPTER 31

31. Supporting Operations and Volume II Consolidation

31.1 Purpose

This chapter defines the minimum integrated requirements for supporting school operations and closes Volume II. Each area operates independently under role-based control while using shared student, staff, finance, communication, document and audit services.

31.2 Library Management

- Catalogue books and physical copies with ISBN, barcode, condition and shelf.
- Issue, renew, return, reserve and mark lost or damaged items.
- Use existing student and staff identities as borrowers.
- Calculate authorized fines and pass financial obligations through Finance.
- Report overdue items, circulation, losses and stock availability.

31.3 Transport Management

- Manage vehicles, drivers, assistants, routes, stops and capacities.
- Record insurance, roadworthy and maintenance expiry.
- Assign enrolled students to pickup and drop-off stops.
- Detect route-capacity exceptions.
- Integrate transport fees with billing and send route-change alerts.

31.4 Health and Sick-Bay Management

- Maintain restricted medical alerts, allergies, conditions and emergency instructions.
- Record visits, complaint, observation, treatment, medication, referral and guardian notification.
- Limit detailed access to authorized health personnel.
- Expose only necessary safety alerts to teachers.
- Track medicine stock and follow-up cases.

31.5 Discipline and Positive Behaviour

- Report incidents with type, severity, description, witnesses and evidence.
- Support investigation, review, action, guardian notification, appeal and follow-up.
- Record commendations, rewards and positive contributions.
- Restrict sensitive cases and avoid permanent stigmatizing labels.
- Report trends and unresolved actions using appropriate confidentiality.

31.6 Inventory and Store Management

- Manage categories, items, units, suppliers, locations and reorder levels.
- Record approved stock in, stock out, transfer, adjustment, damage and loss.
- Calculate quantity from approved transactions.
- Track issues to staff, students or departments.
- Alert on low stock, expiring items and unusual adjustments.

31.7 General Document Management

- Support student, guardian, staff, academic, finance, health, discipline and operational documents.
- Enforce file type, size, secure naming and access rules.
- Maintain category, version, verification, expiry, retention and archive state.
- Prevent executable uploads and provide a malware-scanning integration point.
- Audit view, download, replacement and verification.

31.8 Shared Operational Requirements

Capability	Requirement
Identity	Reuse authoritative student, guardian and staff records
Permissions	Check module, action and record scope on the server
Search	Provide filters, sorting, pagination and safe export
Notifications	Use the common communication service
Documents	Use centralized storage and version controls
Reporting	Support operational and scheduled role-based reports
Audit	Record all sensitive creation, change, approval and export events

31.9 Cross-Module Reporting

Supporting operations contribute weekly, monthly, termly and yearly metrics to responsible officers. Reports focus on overdue work, safety issues, unresolved incidents, stock risk, compliance expiry and assigned corrective actions.

31.10 Security and Testing

Medical and discipline information uses stronger access restrictions. Transport and inventory changes preserve history. Tests cover borrower and route eligibility, bed-free scope is not applicable, stock transaction accuracy, secure files and campus isolation.

31.11 Volume II Completion Gate

- Promotion and JHS completion are controlled and auditable.
- Academic analytics are explainable and role scoped.
- One configurable workflow engine serves approvals.
- Finance supports complete billing-to-reconciliation flow.
- Communication delivers scheduled role-based reporting.
- Supporting operations integrate through shared services.
- No SHS, university, semester, credit-hour or GPA concepts are introduced.

31.12 Conclusion

Volume II completes the Functional Modules and Processing Engines specification for the current PBSMS scope. The next optional volume is the Technical Architecture and Implementation Blueprint: database schema, APIs, service contracts, UI map, security controls, background jobs, migrations and test traceability.

Chapter Control Summary

This chapter is complete when its data ownership, permission boundaries, workflow states, notifications, audit events, reports and acceptance tests are traceable to the technical implementation.

Volume II Traceability Summary

Chapter	Primary domain	Key controlled outcome
22	Academic progression	Approved promotion or JHS completion
23	Academic intelligence	Explainable insight and intervention
24	Governance service	Configurable approval workflow
25-29	Finance	Reconciled student financial position
30	Communication	Role-scoped delivery and follow-up
31	Supporting operations	Integrated daily school operations

Final Acceptance Statement

Volume II is approved when every chapter is represented in the product backlog and traceability matrix; sensitive operations have server-side permission checks; workflows are versioned and auditable; reports reconcile to their source data; and all automated email reporting respects recipient scope.

End of Volume II

PREMIUM BASIC SCHOOL MANAGEMENT SYSTEM

Enterprise Software Requirements Specification

VOLUME III

Technical Architecture and Implementation Blueprint
Chapters 32-45

Nursery - Kindergarten - Primary - Junior High School
Version 1.0 | July 2026

Document Control

Item	Value
Document	PBSMS Enterprise SRS - Volume III
Scope	Technical Architecture and Implementation Blueprint
Chapters	32 through 45
Education levels	Nursery through JHS
Version	1.0
Status	Consolidated specification
Date	July 2026

Purpose

This volume translates the enterprise and functional requirements in Volumes I and II into an implementation-ready technical architecture while protecting the existing Premium Grading System.

Scope Boundary

PBSMS remains a Nursery-to-JHS platform. Senior High School, tertiary semesters, credit hours, GPA and university structures are outside this specification.

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CHAPTER 32

32. Technology Architecture and Existing-System Upgrade Strategy

32.1 Purpose

PBSMS shall be implemented as a safe upgrade of the existing Premium Grading System. Existing working features, records, configurations and access controls shall be inspected and preserved. The target is one modular Nursery-to-JHS platform, not a disconnected replacement.

32.2 Mandatory Baseline Audit

- Identify stack, framework, database, ORM, authentication, roles, routes, services, jobs and tests.
- Classify features as working, partial, broken, missing, duplicated, security risk, migration risk or performance risk.
- Document grading, score entry, result approval, report-card and promotion behavior before changing them.

32.3 Protection Rules

- Do not reset the database or delete history.
- Do not replace the stack solely from preference.
- Use safe migrations for schema changes.
- Preserve user-owned changes and make a recoverable backup.
- Complete only the approved phase and report unresolved risk honestly.

32.4 Layered Architecture

Layer	Responsibility
Presentation	Dashboards, forms, navigation and accessible interaction
Application/API	Requests, validation, authorization and transaction coordination
Domain	Academic, student, finance and workflow rules
Data access	Queries, repositories, transactions and concurrency
Infrastructure	Database, files, email, queues, monitoring and backup

32.5 Shared Services

- Authentication and authorization
- Audit and workflow
- Notification, email and scheduling
- Document generation and storage
- Reporting and numbering
- Integration adapters

32.6 Transactions and Idempotency

Admission conversion, enrolment, transfer, publication, promotion, invoice generation, payment posting, allocation and reversal require explicit transaction boundaries. Retrying sensitive operations must not duplicate records.

32.7 Technical Completion

Each phase shall include migrations, seeds where needed, linting, type checks, automated tests, permission tests, responsive checks, changed-file summary, rollback instructions and known limitations.

Chapter Completion Control

This chapter is complete when its architecture, permissions, data effects, migrations, operational controls and acceptance tests are traceable to implementation evidence.

CHAPTER 33

33. Database Architecture, Schema Standards and Data Migration

33.1 Purpose

The relational database shall preserve existing records while enforcing entity, relationship, domain and business integrity. Physical choices shall follow inspection of the current project.

33.2 Conventions

- Use one naming convention such as plural snake_case tables and singular foreign-key fields.
- Every table has an immutable primary key; business numbers are separate identifiers.
- Use date types for dates, timestamps for events and fixed-precision decimal types for money.
- Store timestamps consistently and display them using the configured school time zone.

33.3 Common Fields

- school_id and campus_id where applicable
- status and effective dates
- created_at, created_by, updated_at and updated_by
- deleted_at, deleted_by and reason where soft deletion is appropriate
- submission and approval fields for controlled workflows

33.4 Integrity

- Foreign keys enforce valid relationships.
- Unique constraints prevent duplicate admission, enrolment, attendance, score, invoice and receipt records.
- Indexes support scope, foreign keys, academic period, status, date and reference searches.
- Protected records use versioning, reversal or archival rather than deletion.

33.5 Principal Table Groups

Domain	Examples
Identity	users, roles, permissions, user_roles, sessions
Student	students, guardians, student_guardians, enrolments, transfers
Academic	years, terms, divisions, levels, classes, subjects, assignments
Assessment	configurations, assessments, scores, grading policies, results
Finance	fee structures, invoices, payments, allocations, reversals, reconciliations
Operations	documents, library, transport, health, discipline, inventory
Governance	workflows, notifications, reports, audit_logs, jobs

33.6 Historical Preservation

Student identity is separate from yearly enrolment; assessment, grading, result and report card are separate; posted payments are immutable; workflow instances retain their definition version.

33.7 Migration

Inventory and profile legacy data, map fields, back up, test migration, validate counts and relationships, reconcile academic and finance totals, approve, execute and verify. Every migration documents rollback.

Chapter Completion Control

This chapter is complete when its architecture, permissions, data effects, migrations, operational controls and acceptance tests are traceable to implementation evidence.

CHAPTER 34

34. Entity-Relationship Model and Complete Table Groups

34.1 Core Model

The central relationship is Student -> Enrolment -> academic and operational transactions. This retains permanent identity while preserving class and academic-year context.

34.2 Institutional Relationships

- One school has many campuses, users, students, staff and academic years.
- Users may have several roles and scope assignments.
- A staff, guardian or student profile may link to one user account.
- Configuration records remain scoped and historically versioned.

34.3 Academic Relationships

- Academic year has terms and yearly classes.
- Division has class levels; class level may point to a next level.
- Class has enrolments, class subjects, teacher assignments and timetable entries.
- Teacher access derives from active assignment.

34.4 Student Lifecycle

- Admission remains separate until approved conversion.
- One student has many historical enrolments and guardian relationships.
- Transfer links source and destination enrolments.
- Promotion links current enrolment, policy, annual performance, decision and new enrolment.
- JHS completion links the final enrolment, approval and issued certificate.

34.5 Academic Processing

Assessment has many student scores; grading references policy versions; result has subject outcomes and versions; report card references a result version and template version.

34.6 Finance Relationships

Fee structures generate invoice items; payments connect to invoices through allocations; receipts reference posted payments; reversals preserve the original; reconciliation contains matches and variances.

34.7 Workflow and Communication

Workflow definitions have versions, stages and conditions; instances have tasks and decisions. Notifications separate message from recipients, attempts, acknowledgements and actions.

34.8 Prohibited Relationships

- No score without valid assessment and enrolment.
- No report card without approved result version.
- No allocation without posted payment and invoice.
- No guardian access without active relationship.
- No teacher score entry outside assignment.
- No cross-school relationship unless explicitly authorized.

Chapter Completion Control

This chapter is complete when its architecture, permissions, data effects, migrations, operational controls and acceptance tests are traceable to implementation evidence.

CHAPTER 35

35. API Architecture and Service Contracts

35.1 Purpose

APIs expose authorized business capabilities, never unrestricted database access. Internal and external consumers use stable contracts with server-side validation, permissions, scope and audit.

35.2 Standards

- Versioned base path for formal APIs
- Plural resource names and explicit state-transition actions
- Consistent JSON data, metadata and error structures
- Request and correlation identifiers
- Documented status codes and machine-readable errors

35.3 Security

Every protected request checks authentication, account state, permission, organizational scope, record relationship and valid business state. Guardian, student, teacher, finance and health access receive dedicated record-level checks.

35.4 Collections

Large collections use server-side pagination. Filter and sort fields are allowlisted. Search returns only authorized information. Nested expansions are controlled.

35.5 Idempotency and Concurrency

Admission conversion, promotion, invoice, payment, callback, receipt, report and document generation require idempotency. Version values or locks prevent lost updates.

35.6 Domain Contracts

- Identity and configuration
- Admissions, students, guardians and enrolments
- Attendance, assessment, grading, results and report cards
- Promotion and JHS completion
- Fees, invoices, payments, assistance and reconciliation
- Workflow, communication, documents and supporting operations

35.7 Webhooks

Incoming webhooks verify provider, signature, timestamp, replay window, event ID, reference, schema, amount and currency. Longer processing moves to idempotent background jobs.

35.8 Testing

Contract, permission, finance, academic, webhook, performance and version-compatibility tests are mandatory. Hidden interface controls do not prove authorization.

Chapter Completion Control

This chapter is complete when its architecture, permissions, data effects, migrations, operational controls and acceptance tests are traceable to implementation evidence.

CHAPTER 36

36. Application Security, Privacy and Access-Control Architecture

36.1 Security Model

PBSMS applies least privilege, defense in depth, explicit authorization, secure defaults, data minimization, segregation of duties, accountability and privacy by design.

36.2 Authentication

- Unique named accounts
- Adaptive password hashing such as Argon2id or bcrypt
- Rate-limited login and secure reset
- Optional or required MFA for privileged users
- Secure session cookies, rotation, timeout and revocation
- No shared operational accounts

36.3 Authorization

Effective access combines active role, permission, scope, record relationship, record state, workflow authority and delegation. View, edit, approve, publish, reverse and export remain separate.

36.4 Segregation

- Cashier cannot approve own reversal.
- Score entry is separate from final publication.
- High-privilege role assignment receives extra review.
- Backup creation and production restoration are distinct authorities.

36.5 Data Protection

Class	Examples
Public	Approved announcements
Internal	Timetables and assignments
Confidential	Student, guardian, results and statements
Highly Confidential	Medical, bank, salary, IDs and credentials

36.6 Application Controls

- Parameterized database queries
- Output escaping and safe rich-text sanitization
- CSRF and secure headers
- File signature validation and private storage
- Protection from open redirects, SSRF, injection and path traversal
- Secret management outside source control

36.7 Monitoring and Privacy

Security and business audit events are append-only for ordinary users. Children, medical and discipline data use stricter minimization, retention, export and email rules.

36.8 Incident Response

Detect, classify, contain, preserve evidence, investigate, recover, notify authorized parties, correct root cause and monitor. Security tests cover identity, scope, files, privacy and integrations.

Chapter Completion Control

This chapter is complete when its architecture, permissions, data effects, migrations, operational controls and acceptance tests are traceable to implementation evidence.

CHAPTER 37

37. Frontend Architecture, User Experience and Navigation System

37.1 Purpose

The frontend shall be consistent, responsive, accessible and role-sensitive while preserving familiar existing grading-system patterns where suitable.

37.2 Application Shell

- Header with school, context, search, notification and user menu
- Permission-generated collapsible sidebar
- Main content with breadcrumbs
- Responsive drawer on small screens
- Visible active academic year, term and campus

37.3 Role Dashboards

Executives see institutional summaries; headmasters see approvals and risks; coordinators see academic completion; teachers see assigned work; accountants see collections and exceptions; parents and students see only their own authorized information.

37.4 Design System

Central tokens govern colors, typography, spacing and breakpoints. Reusable buttons, inputs, tables, cards, dialogs, alerts, timelines and status badges prevent fragmented interfaces.

37.5 Forms and Tables

Forms use visible labels, logical sections, conditional fields, draft states and server-backed validation. Tables use server-side search, filters, sorting, pagination, safe bulk actions and meaningful empty states.

37.6 Workflow UX

Every controlled record shows current stage, completed stages, pending owner, deadline, comments, return reason and escalation. Approval screens expose evidence and consequences.

37.7 Mobile and Accessibility

Critical attendance, score, notification, approval and parent functions remain practical on mobile. Keyboard access, screen-reader labels, contrast, focus, zoom and reduced motion are tested.

37.8 State and Security

Loading, success, error, offline, empty and restricted states are distinct. The frontend hides unavailable actions, but the server remains authoritative.

Chapter Completion Control

This chapter is complete when its architecture, permissions, data effects, migrations, operational controls and acceptance tests are traceable to implementation evidence.

CHAPTER 38

38. Background Jobs, Automation, Scheduling and Event Processing

38.1 Purpose

Scheduled, resource-intensive and retryable work shall run through observable background jobs. Automation prepares, calculates and communicates but remains subordinate to human approval for sensitive decisions.

38.2 Job Model

- Job type, owner, scope and related record
- Parameters, idempotency key and priority
- Scheduled, started, completed and failure times
- Attempts, progress, error and result
- Statuses from queued through completed, retry, failed and dead letter

38.3 Queues and Workers

Separate queues may serve critical, finance, academic, documents, reports, communication, maintenance and backup work. Workers claim atomically, heartbeat, recover abandoned jobs and use least privilege.

38.4 Retry and Recovery

Temporary failures use bounded backoff; permanent validation failures do not retry automatically; dead-letter items remain visible; partial batches preserve item outcomes and retry only failures.

38.5 Scheduler

Schedules support one-time, daily, weekly, monthly, termly, yearly and event-triggered execution using the school time zone. Missed-run policy is explicit.

38.6 Events

Versioned domain events such as StudentEnrolled, ResultPublished and PaymentPosted are published after commit. Consumers are idempotent; transactional outbox and consumption logs prevent loss and duplication.

38.7 Automation Coverage

Attendance registers and alerts, assessment reminders, grading, result compilation, report cards, promotion recommendations, billing, overdue status, payment callbacks, reconciliation, reports, emails, actions, expiry checks and backups.

38.8 Monitoring

Operations dashboards show queue depth, running and failed jobs, retries, worker health, missed reports and last backup. Tests simulate worker, queue, provider and database failures.

Chapter Completion Control

This chapter is complete when its architecture, permissions, data effects, migrations, operational controls and acceptance tests are traceable to implementation evidence.

CHAPTER 39

39. Reporting, Analytics and Data-Projection Architecture

39.1 Purpose

Reports convert authorized operational data into accurate measures, trends, recommendations and accountable actions. Every figure has a source, formula, period, scope and freshness.

39.2 Report Types

- Operational reports for immediate work
- Management reports for periodic performance
- Executive summaries for institutional decisions
- Compliance and audit reports
- Personal student, guardian and staff reports

39.3 Definitions and Projections

Every report has owner, version, parameters, permission, scope, formats and sensitivity. Reporting projections document their source, refresh trigger, freshness, rebuild and reconciliation.

39.4 KPI Model

KPI definitions contain formula, source, target, weight, thresholds, frequency, owner, scope and version. Calculations retain numerator, denominator, actual, target, status and exceptions.

39.5 Domain Reporting

- Academic, attendance and promotion
- Billing, collections, outstanding fees and reconciliation
- Admissions and staff
- Library, transport, health, discipline and inventory
- Workflow completion, actions and escalations

39.6 Dashboards and Exports

Widgets declare dataset, permission, scope, refresh and drill-down. PDF, spreadsheet and CSV exports are controlled, audited, protected from formula injection and moved to jobs when large.

39.7 Scheduled Delivery

Weekly, monthly, termly and yearly reports resolve current authorized recipients, use secure links for sensitive data, track delivery and acknowledgement, and create follow-up actions.

39.8 Reconciliation

Academic summaries reconcile to approved result versions; financial reports reconcile invoices, assistance, payments, credits, reversals, adjustments and write-offs. Permission, calculation and delivery tests are mandatory.

Chapter Completion Control

This chapter is complete when its architecture, permissions, data effects, migrations, operational controls and acceptance tests are traceable to implementation evidence.

CHAPTER 40

40. File Storage, Document Generation and Digital Verification Architecture

40.1 Purpose

Files and official documents use centralized storage, metadata, authorization, versioning, retention and audit services rather than module-specific public paths.

40.2 Storage and Metadata

The storage abstraction supports private local, network or object storage. Metadata records original and stored names, media type, size, checksum, provider, storage key, classification, version, uploader and retention.

40.3 Secure Upload

Permission, size, extension, detected signature, temporary storage and malware scan precede activation. Executables, double-extension tricks, unsafe paths and mismatched content are rejected.

40.4 Access

Document retrieval checks identity, permission, owning-record access, school, campus, classification, status, version and expiry. Protected content streams through authorization or short-lived signed links.

40.5 Versioning

Official report cards, certificates, letters, statements and templates create new versions rather than overwrite history. Each version stores checksum, predecessor, reason, creator and current state.

40.6 Document Engine

Approved source snapshot -> template version -> render -> visual and technical validation -> checksum -> private storage -> verification record -> authorized distribution.

40.7 Signatures and Verification

Signatures and seals are restricted and versioned. QR codes contain a random verification reference, never confidential details. Public verification shows minimal authenticity and status.

40.8 Retention and Testing

Documents follow category retention, archival, backup and secure disposal. Tests cover upload attacks, authorization, long content, batch generation, versioning, QR readability, checksums and restoration.

Chapter Completion Control

This chapter is complete when its architecture, permissions, data effects, migrations, operational controls and acceptance tests are traceable to implementation evidence.

CHAPTER 41

41. External Integrations and Provider-Adapter Architecture

41.1 Purpose

External capabilities connect through vendor adapters while PBSMS remains authoritative for identity, permission, academic history, financial truth and audit.

41.2 Registry and Credentials

Each integration has type, provider, environment, owner, health, configuration reference and credential reference. Secrets remain outside code, are scoped, rotatable and separately managed for sandbox and production.

41.3 Payments

PBSMS creates an internal reference; callbacks validate provider, signature, time, replay, reference, amount, currency and event ID; duplicate delivery is idempotent; provider status maps to controlled PBSMS state; settlement is reconciled.

41.4 Communication

Email and SMS adapters expose stable internal send and status contracts. Bounce, failure, retry, consent, confidentiality and cost thresholds remain centralized.

41.5 Devices and Education Systems

Biometric devices are registered and mapped to PBSMS identities; offline events retain original time and unique IDs. Learning, accounting and government connectors have explicit field ownership and mappings.

41.6 Synchronization

Direction, source of truth, cursor, scope, archive behavior and conflict rules are documented. External deletion does not remove protected PBSMS history.

41.7 Reliability

Timeouts, circuit breakers, bounded retries, exception queues and graceful degradation prevent false success and protect core operations.

41.8 Monitoring and Testing

Integration dashboards show health, last success, failures, queue, latency and credential expiry. Adapter, payment, communication, device, synchronization and security tests are required.

Chapter Completion Control

This chapter is complete when its architecture, permissions, data effects, migrations, operational controls and acceptance tests are traceable to implementation evidence.

CHAPTER 42

42. Performance, Scalability, Reliability and Observability Architecture

42.1 Targets

Standard reads and writes should complete within measured targets under normal load; large reports are accepted quickly as jobs. Final targets follow baseline measurement and realistic school workloads.

42.2 Capacity and Peaks

Plan for students, guardians, staff, concurrent users, attendance, scores, invoices, documents and audit growth. Simulate morning attendance, score deadlines, results, term opening and payment deadlines.

42.3 Scaling

Stateless application instances, shared sessions, database, queues and storage allow horizontal scaling. Web, worker, report, document and integration capacity can scale independently.

42.4 Database and Cache

Use query plans, pagination, indexes, bounded connection pools, short transactions and reporting projections. Cache keys preserve school, campus, user scope, parameters and version; caches never bypass authorization.

42.5 Concurrency

Optimistic versions, short locks, unique constraints, atomic updates and idempotency prevent lost updates and duplicates for scores, invoices, allocations, receipt numbers, promotion and inventory.

42.6 Reliability

Define availability, RTO and RPO. Health checks distinguish liveness and readiness. Nonessential provider failure degrades gracefully without undoing valid business transactions.

42.7 Observability

Structured logs, metrics and traces use correlation IDs while excluding secrets. Monitor latency, errors, database, queues, workers, storage, integrations, backup and business-process health.

42.8 Alerts and Testing

Actionable alerts have severity, owner and escalation. Load, stress, endurance, failure, concurrency and recovery tests prove capacity and safe failure behavior.

Chapter Completion Control

This chapter is complete when its architecture, permissions, data effects, migrations, operational controls and acceptance tests are traceable to implementation evidence.

CHAPTER 43

43. Backup, Disaster Recovery and Business Continuity Architecture

43.1 Purpose

Backups protect database, files, configuration, application releases and recovery material against failure, error, attack and facility loss. A backup is trusted only after verification and restoration testing.

43.2 Recovery Objectives

Classify critical functions, maximum tolerable outage, RTO, RPO, dependencies, manual procedures and owners. Authentication, student identity, finance, security and emergency communication receive high priority.

43.3 Scope and Schedule

Back up database, uploaded and generated files, policies, workflows, templates, integration configuration and released application versions. Use frequent incremental or log backup, daily verified backup, weekly full and retained monthly points according to policy.

43.4 Storage Security

Maintain live data plus local/nearest and off-site copies; use encryption, restricted access, checksums, inventories, retention and, where supported, immutability.

43.5 Verification and Restore

Verify archive readability, counts, relationships, files, checksums and migration state. Restore to isolation, match application version, reconcile database and files, then verify identity, academics, finance, documents, jobs and integrations.

43.6 Disaster Scenarios

Document response for hardware failure, corruption, faulty migration, ransomware, account compromise, facility loss, internet outage and provider outage.

43.7 Continuity

Controlled manual attendance, payment and academic procedures identify original event time, responsible user, supporting evidence, later re-entry and reconciliation.

43.8 Governance and Testing

Named incident, technical, database, finance, academic, communication and security roles coordinate recovery. Restoration and continuity exercises record actual recovery time, issues and corrective actions.

Chapter Completion Control

This chapter is complete when its architecture, permissions, data effects, migrations, operational controls and acceptance tests are traceable to implementation evidence.

CHAPTER 44

44. Testing, Quality Assurance and Requirements Traceability

44.1 Quality Model

A feature is complete only when its database, backend, frontend, permissions, validation, audit, notifications, reports, tests, documentation and acceptance criteria are satisfied.

44.2 Test Levels

Unit, component, integration, API contract, system, end-to-end and user-acceptance testing use isolated environments and synthetic Nursery-to-JHS data.

44.3 Functional Coverage

Test authentication, roles, admissions, students, academic structure, attendance, assessment, grading, results, report cards, promotion, finance, communication and supporting operations.

44.4 Security and Migration

Attempt injection, broken authorization, cross-campus access, unsafe files and webhook replay. Migration tests compare counts, identifiers, relationships, scores, results, invoice balances, payments and documents.

44.5 Performance and Reliability

Load, stress, endurance, concurrency, dependency-failure and recovery tests reflect morning attendance, score deadlines, result access, billing and report generation.

44.6 Regression

Baseline the existing grading behavior and rerun after relevant changes. Existing scores, calculations, published results, report cards and roles remain protected.

44.7 Defects

Severity	Meaning
Critical	Security, financial integrity, data loss or total outage
High	Major workflow unavailable or official result incorrect
Medium	Important impairment with workaround
Low	Minor usability or presentation issue

44.8 Traceability

Every stable requirement ID maps to chapter, design, implementation, database, API, permission, tests, status and evidence. Critical defects block release; role-based user acceptance is recorded.

Chapter Completion Control

This chapter is complete when its architecture, permissions, data effects, migrations, operational controls and acceptance tests are traceable to implementation evidence.

CHAPTER 45

45. Deployment, Environment and Release-Management Architecture

45.1 Purpose

Deployment coordinates approved application artifacts, database migrations, configuration, workers, schedules, storage, integrations, monitoring, backup and rollback.

45.2 Environments

Local, development, automated test, staging and production use separate databases, storage, secrets, queues, providers, logs and backups. Staging resembles production; nonproduction outputs are visibly marked.

45.3 Source and Build

Repositories contain source, migrations, tests, templates, build definitions and lock files, never secrets or backups. Reproducible artifacts record source commit, runtime, dependency checksum, migration version and artifact checksum.

45.4 Migration and Pipeline

Backup and prechecks precede additive migrations, backfills and validation. Pipeline: review -> tests -> build -> staging -> migration test -> acceptance -> approval -> backup -> production -> smoke test -> monitoring.

45.5 Release Strategy

In-place, rolling, blue-green or canary deployment may be selected. Feature flags support gradual rollout but retain permission enforcement and have owners and removal plans.

45.6 Rollback

Rollback triggers include authentication, authorization, academic calculation, financial integrity, migration and severe performance failures. Preserve valid post-release transactions; corrective forward migration may be safer than restoration.

45.7 Verification and Handover

Verify login, permissions, student, attendance, scores, results, finance, documents, workers, schedules, integrations and audit. Provide architecture, deployment, configuration, recovery, monitoring, administration and support documentation.

45.8 Volume III Completion

Technical architecture is complete when it is traceable to functional requirements; migrations preserve history; APIs enforce scope; automation recovers safely; reports reconcile; backup restore is tested; and deployment and rollback are repeatable.

Chapter Completion Control

This chapter is complete when its architecture, permissions, data effects, migrations, operational controls and acceptance tests are traceable to implementation evidence.

Volume III Traceability Summary

Chapter	Technical area	Controlled outcome
32	Architecture	Safe upgrade and module boundaries
33-34	Data	Integrity, history and relationships
35-36	API and security	Authorized service access
37	Frontend	Role-based usable interface
38-39	Automation and reporting	Recoverable processing and reconciled insight
40-41	Documents and integrations	Secure external capabilities
42-43	Reliability and recovery	Measured service and tested restoration
44-45	Quality and release	Traceable, repeatable deployment

Final Acceptance Statement

Volume III is approved when technical designs are mapped to the functional requirements; existing grading history is preserved; server-side permissions and data scope are enforced; migrations and rollback are tested; reports reconcile; backups restore; and production releases are repeatable and observable.

End of Volume III

PREMIUM BASIC SCHOOL MANAGEMENT SYSTEM

Enterprise Software Requirements Specification

VOLUME IV

Codex Implementation Blueprint
Chapters 46-55

Nursery - Kindergarten - Primary - Junior High School
Version 1.0 | July 2026

Document Control

Item	Value
Document	PBSMS Enterprise SRS - Volume IV
Scope	Codex Implementation Blueprint
Chapters	46 through 55
Education levels	Nursery through JHS
Currency	Ghana cedi (GHS)
Version	1.0
Status	Consolidated implementation specification
Date	July 2026

Purpose

This volume converts the enterprise, functional and technical requirements in Volumes I-III into a controlled sequence of prompts that Codex can execute against the existing Premium Grading System.

Execution Rule

Use one numbered prompt at a time. Codex shall inspect first, preserve existing data and functionality, implement only the current scope, verify it, report evidence and stop before the next prompt.

Scope Boundary

PBSMS serves Nursery through Junior High School. Senior High School, university structures, semesters, credit hours, GPA and CGPA remain outside scope.

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CHAPTER 46

46. Codex Master Execution Instructions

CODEX EXECUTION PROMPT

Apply the Volume IV master rules. Inspect the existing implementation before changing it, preserve all working behavior and historical records, complete only this chapter's current subprompt, run proportionate verification, report evidence and stop.

46.1 Purpose

These instructions govern every Codex implementation phase. PBSMS is an upgrade of the existing Premium Grading System, not a new or separate project. Codex shall preserve the working stack, data, accounts, grading logic and institutional history.

46.2 Mandatory First Action

- Inspect the complete repository, database model, authentication, roles, routes, services, background jobs, design system and tests.
- Record the project root, technology stack, framework versions, database engine, working modules, incomplete features and known risks.
- Review source-control status and confirm a recoverable backup before material changes.
- Prepare a short plan for only the current numbered prompt.

46.3 Protection Rules

- Do not create another project folder or rebuild from scratch without explicit authority.
- Do not delete existing data, reset the database, remove working features or overwrite historical results.
- Do not rename fields without safe migration and rollback instructions.
- Do not bypass server-side permissions or use interface hiding as authorization.
- Do not continue automatically to another numbered prompt.

46.4 Implementation Standard

Every completed module must include its database changes, backend services, role permissions, validation, interface, notifications, audit events, reports, tests, migration notes and integration evidence.

46.5 Common Architecture

- One shared platform with controlled module boundaries and data ownership.
- Student identity remains separate from yearly enrolment.
- Users may hold several scoped roles.
- Sensitive decisions use configurable approval workflows.
- Automation handles repetitive work but never silently makes irreversible institutional decisions.

46.6 Completion Gate

- Run safe migrations and seeds where required.
- Run formatting, linting, type checks, builds and automated tests.
- Test permission and record-scope boundaries.
- Summarize changed files, migrations, rollback steps and unresolved risks.
- Stop after the current prompt.

Chapter Completion Control

This chapter is complete only when its implementation evidence, data effects, permissions, tests, migration or rollback information and unresolved risks have been reported honestly.

CHAPTER 47

47. Existing-System Audit and Upgrade Baseline Prompt

47.1 Objective

Create an evidence-based baseline of the Premium Grading System before enterprise development begins. The baseline protects working behavior and prevents assumptions from becoming destructive changes.

47.2 Inventory Prompt

- Catalogue pages, modules, routes, APIs, server actions, database tables, models, services, scheduled jobs and integrations.
- Identify authentication, sessions, roles, permissions, record scopes and audit behavior.
- Document score entry, assessment weighting, grading, approval, publication, report-card and promotion logic.
- Identify file storage, email, exports, backups and deployment configuration.

47.3 Classification

- Working and must be preserved
- Partially working
- Broken
- Missing
- Duplicated or abandoned
- Security risk
- Data-migration risk
- Performance risk

47.4 Required Evidence

Every finding shall reference the relevant file, route, table, screen or reproducible behavior. Codex shall distinguish confirmed facts from assumptions and shall not implement major features during the audit.

47.5 Baseline Tests

- Authentication and account state
- Teacher assignments and score entry
- Existing grade calculations
- Result publication
- Report-card generation
- Historical academic records
- Current permission boundaries

47.6 Deliverables

- System inventory
- Data and relationship map
- Working-feature preservation list
- Risk register
- Regression baseline
- Phased upgrade recommendation

47.7 Acceptance

The audit is complete only when existing behavior is reproducible, risks are prioritized, critical data is backed up and the next implementation phase can begin without guessing.

Chapter Completion Control

This chapter is complete only when its implementation evidence, data effects, permissions, tests, migration or rollback information and unresolved risks have been reported honestly.

CHAPTER 48

48. Phased Implementation Backlog and Dependency Map

48.1 Objective

Translate the SRS into an ordered backlog that respects technical and business dependencies. Work proceeds in small, testable phases rather than attempting the entire platform at once.

48.2 Delivery Sequence

- Foundation: architecture, database standards, identity, roles, audit and configuration.
- Academic setup: years, terms, divisions, levels, classes, subjects and assignments.
- Student lifecycle: admissions, guardians, enrolment, transfer, promotion and completion.
- Academic processing: attendance, assessments, grading, results and report cards.
- Finance: fee structures, invoices, assistance, payments, receipts and reconciliation.
- Operations: timetable, communication, library, transport, health, discipline, inventory and documents.
- Intelligence: scheduled reports, KPIs, actions, escalation and executive dashboards.
- Technical completion: security, quality assurance, migration, deployment and handover.

48.3 Backlog Item Structure

- Stable requirement and task identifier
- Objective and business value
- Dependencies and affected modules
- Database, backend and frontend work
- Permissions and scope
- Audit and notification events
- Migration and rollback
- Tests and acceptance criteria

48.4 Dependency Controls

A phase shall not consume records or services that do not yet have an approved owner and contract. Shared master data, permissions, audit and configuration precede dependent modules.

48.5 Definition of Ready

A task is ready when its scope, dependencies, data effects, roles, business rules and acceptance tests are known.

48.6 Definition of Done

A task is done only when implementation, migrations, permissions, validation, integration, audit, tests, documentation and review evidence are complete.

48.7 Change Control

New requirements are assessed for scope, dependency, migration, security, cost and schedule impact before entering the ordered backlog.

Chapter Completion Control

This chapter is complete only when its implementation evidence, data effects, permissions, tests, migration or rollback information and unresolved risks have been reported honestly.

CHAPTER 49

49. Database Migration and Data-Preservation Prompt Pack

CODEX EXECUTION PROMPT

Apply the Volume IV master rules. Inspect the existing implementation before changing it, preserve all working behavior and historical records, complete only this chapter's current subprompt, run proportionate verification, report evidence and stop.

49.1 Objective

Upgrade the database safely while preserving users, students, staff, classes, subjects, scores, results, report cards and all other protected history.

49.2 Schema Audit Prompt

- Profile every existing table, column, type, null pattern, index, constraint and relationship.
- Identify duplicate identities, orphan records, invalid status values and inconsistent dates.
- Record table size, growth and high-risk transformations.

49.3 Target Schema Prompt

- Separate permanent student identity from academic-year enrolment.
- Separate assessment, score, grading policy, result version and report-card version.
- Use scoped roles and effective dates.
- Use immutable financial postings and explicit reversals.
- Add audit fields, foreign keys, unique constraints and indexes safely.

49.4 Migration Design Prompt

- Prefer additive migrations and phased backfills.
- Document source-to-target field mappings.
- Resolve ambiguous records through review queues rather than guesswork.
- Use transactions and checkpoints for critical transformations.
- Provide forward and rollback procedures.

49.5 Preservation Tests

- Compare pre- and post-migration record counts.
- Reconcile identifiers, relationships and historical periods.
- Recalculate representative existing results without changing approved values.
- Verify existing accounts and permission mappings.
- Confirm documents remain connected to their records.

49.6 Financial Integrity

Use fixed-precision decimal values, retain original payments, represent corrections through reversals or adjustments, and prove invoice, allocation, credit and balance reconciliation.

49.7 Migration Acceptance

No production migration proceeds without a verified backup, rehearsed migration, reconciliation report, rollback plan and authorized approval.

Chapter Completion Control

This chapter is complete only when its implementation evidence, data effects, permissions, tests, migration or rollback information and unresolved risks have been reported honestly.

CHAPTER 50

50. Backend, API and Domain-Service Implementation Prompt Pack

CODEX EXECUTION PROMPT

Apply the Volume IV master rules. Inspect the existing implementation before changing it, preserve all working behavior and historical records, complete only this chapter's current subprompt, run proportionate verification, report evidence and stop.

50.1 Objective

Implement business capabilities through modular domain services and secure APIs while keeping the existing stack and patterns where they remain sound.

50.2 Service Boundaries

- Identity and access
- School and academic configuration
- Admissions and student lifecycle
- Attendance and academic processing
- Results, report cards and promotion
- Finance
- Communication and reporting
- Supporting operations and governance

50.3 Service Rules

- Domain services enforce business rules independently of the interface.
- Every protected operation checks identity, permission, scope, record relationship and workflow state.
- Critical operations define transaction and idempotency boundaries.
- Errors use stable codes and safe user messages.
- Sensitive changes emit audit events.

50.4 API Prompt

- Use consistent versioned contracts, validation, pagination, filtering and safe errors.
- Prevent mass assignment and unrestricted relationship expansion.
- Require idempotency for admission conversion, promotion, invoices, payments and document generation.
- Use optimistic concurrency or controlled locks for competing edits.
- Generate contract and permission tests.

50.5 Workflow Prompt

Implement configurable stages, actors, conditions, deadlines, decisions, return reasons, delegation, escalation and completion actions. Workflow versions remain attached to historical instances.

50.6 Event and Job Prompt

Publish domain events after successful commit. Consumers shall be idempotent, observable and recoverable. Long-running reports, documents, emails, batches and integrations use controlled jobs.

50.7 Acceptance

Backend completion requires passing domain, API, integration, concurrency, authorization and audit tests with documented migrations and rollback.

Chapter Completion Control

This chapter is complete only when its implementation evidence, data effects, permissions, tests, migration or rollback information and unresolved risks have been reported honestly.

CHAPTER 51

51. Frontend, Role-Based Dashboard and User-Experience Prompt Pack

CODEX EXECUTION PROMPT

Apply the Volume IV master rules. Inspect the existing implementation before changing it, preserve all working behavior and historical records, complete only this chapter's current subprompt, run proportionate verification, report evidence and stop.

51.1 Objective

Implement a responsive and accessible interface that shows each user the work, records and decisions permitted by their role and scope.

51.2 Application Shell

- School and campus identity
- Active academic year and term
- Permission-generated collapsible navigation
- Breadcrumbs, global search, notifications and user menu
- Responsive drawer for mobile devices

51.3 Role Dashboards

- Executives: institutional KPIs, risks and actions.
- Headmaster: academic, attendance, staff, finance and approvals.
- Academic coordinator: assignments, assessment completion and result workflow.
- Teacher: assigned classes, subjects, attendance and score tasks.
- Accountant: collections, balances, reconciliation and exceptions.
- Parent and student: only linked or personal authorized records.

51.4 Page Standard

- Search, filters, sorting and pagination
- Loading, empty, error, restricted and success states
- Accessible forms and validation messages
- Responsive tables and safe bulk actions
- Status, workflow timeline and audit history
- Print and export only where permitted

51.5 Workflow Experience

Controlled records show current stage, previous decisions, pending owner, deadline, return reason and permitted actions. Approval dialogs show consequences before confirmation.

51.6 Accessibility

Test keyboard navigation, focus visibility, labels, heading order, contrast, screen readers, zoom, error association and mobile interaction.

51.7 Security Boundary

The interface may hide unavailable actions for clarity, but server authorization remains mandatory. Direct URLs and changed request data must not bypass scope.

51.8 Acceptance

Each standard role completes its primary tasks on mobile and desktop without seeing unauthorized records or controls.

Chapter Completion Control

This chapter is complete only when its implementation evidence, data effects, permissions, tests, migration or rollback information and unresolved risks have been reported honestly.

CHAPTER 52

52. Automation, Reporting, Email and Performance-Monitoring Prompt Pack

CODEX EXECUTION PROMPT

Apply the Volume IV master rules. Inspect the existing implementation before changing it, preserve all working behavior and historical records, complete only this chapter's current subprompt, run proportionate verification, report evidence and stop.

52.1 Objective

Convert operational data into scheduled, role-scoped reports and accountable follow-up while preventing notification overload and sensitive-data exposure.

52.2 Reporting Engine Prompt

- Support daily, weekly, monthly, termly, yearly, event-triggered and manual reports.
- Define report owner, audience, scope, sections, formulas, targets, comparison periods, sensitivity and formats.
- Reconcile academic and financial reports to authoritative transactions and approved versions.
- Generate large reports through background jobs.

52.3 Scheduling Prompt

- Configure delivery day, time, time zone, academic trigger, recipients, CC rules, secure link, attachment policy and escalation.
- Prevent duplicate execution through idempotency.
- Recover missed schedules according to explicit policy.
- Track queued, generated, sent, failed, retried, opened where appropriate and acknowledged.

52.4 Role Reports

- Headmaster: academic performance, attendance, staff, finance, approvals and risk.
- Academic coordinator: assessment completion, missing scores, syllabus coverage and interventions.
- Accountant: collections, outstanding fees, reversals, reconciliation and targets.
- Teachers: assigned work, missing scores, returned submissions and at-risk students.
- Admissions and operational officers: workload, exceptions, deadlines and performance.

52.5 KPI Engine

Every KPI defines source, formula, period, target, weight, thresholds, scope, responsible supervisor and version. Results retain evidence and exceptions.

52.6 Action and Escalation

Reports identify issue, impact, recommended action, responsible person, priority and deadline. Recipients acknowledge, comment, assign, attach evidence, resolve, reopen or escalate.

52.7 Email Privacy

Email contains the minimum necessary detail. Sensitive content is accessed through authenticated, expiring portal links. Failures retry safely and repeated failure alerts administrators.

52.8 Acceptance

Calculations reconcile, recipients match current authority, schedules execute once, emails respect privacy, unresolved actions carry forward and all delivery and decisions are audited.

Chapter Completion Control

This chapter is complete only when its implementation evidence, data effects, permissions, tests, migration or rollback information and unresolved risks have been reported honestly.

CHAPTER 53

53. Security, Privacy and Permission-Verification Prompt Pack

CODEX EXECUTION PROMPT

Apply the Volume IV master rules. Inspect the existing implementation before changing it, preserve all working behavior and historical records, complete only this chapter's current subprompt, run proportionate verification, report evidence and stop.

53.1 Security Baseline Prompt

- Assess authentication, sessions, authorization, files, APIs, logs, secrets, dependencies, backups and integrations.
- Create an evidence-based threat model covering account takeover, privilege escalation, record leakage, result tampering, financial fraud and malicious uploads.
- Classify findings as critical, high, medium, low or informational.

53.2 Authentication and Session Prompt

- Use secure password hashing, single-use expiring reset tokens, rate limiting, lockout and account-state enforcement.
- Rotate sessions after login and privilege change; support inactivity timeout, revocation and logout from all devices.
- Protect state-changing requests with CSRF and origin controls where appropriate.

53.3 RBAC and Scope Prompt

- Evaluate roles, effective dates, delegation, school, campus, division, department, class, subject, ownership, action and workflow stage.
- Deny by default and enforce authorization on the server.
- Test every standard role against every major module.

53.4 Record Isolation Prompt

- Guardians access only actively linked students.
- Students access only their own authorized records.
- Teachers access only assigned classes and subjects.
- Campus users cannot cross campus through changed URLs, APIs, exports or searches.
- Medical and finance roles remain limited to their authorized domains.

53.5 Segregation of Duties

Block configured self-approval and conflicting authority for admissions, results, publication, payment reversal, scholarships, role escalation, restore and sensitive export. Delegation is scoped, dated and audited.

53.6 Privacy and Files

Classify data, mask sensitive fields, restrict exports, redact logs and emails, validate file signatures and sizes, reject executables, store privately and permission-check every download.

53.7 API and Audit Security

Authenticate and scope integrations, verify webhooks, prevent replay and duplicate payments, protect secrets, keep audit records append-only for ordinary users and monitor suspicious activity.

53.8 Incident Response

Detect, classify, contain, investigate, recover, review and close security incidents with restricted evidence and audited emergency actions.

53.9 Final Security Acceptance

Acceptance fails for missing server authorization, cross-record leakage, public confidential files, silent result or payment changes, editable audit history, unverified payment webhooks, committed secrets or unresolved critical vulnerabilities.

Chapter Completion Control

This chapter is complete only when its implementation evidence, data effects, permissions, tests, migration or rollback information and unresolved risks have been reported honestly.

CHAPTER 54

54. Testing, Quality Assurance and Acceptance Prompt Pack

CODEX EXECUTION PROMPT

Apply the Volume IV master rules. Inspect the existing implementation before changing it, preserve all working behavior and historical records, complete only this chapter's current subprompt, run proportionate verification, report evidence and stop.

54.1 Test Strategy Prompt

Create a risk-based strategy and traceability matrix connecting every critical requirement to business rules, roles, test cases, expected results, automation and acceptance status.

54.2 Static and Unit Testing

- Run formatting, linting, type checking, compilation and production build.
- Test academic weighting, grading boundaries, rounding, attendance percentages, promotion, GHS finance calculations and KPI formulas.
- Add regression tests for every repaired defect.

54.3 Database and Integration Testing

- Verify unique constraints, foreign keys, one active enrolment, duplicate prevention and historical preservation.
- Force failures inside admission, publication, promotion, payment and inventory transactions and verify rollback.
- Test full cross-module flows and resulting audit records.

54.4 Permission and Workflow Testing

Exercise every role, action and record scope, including delegation, self-approval prevention, parent isolation, teacher assignments and campus boundaries.

54.5 Functional Acceptance

- Test foundation, student lifecycle, academic operations, finance, supporting operations and governance.
- Verify optional disabled modules do not break the platform.
- Confirm search, filters, pagination, print and export respect scope.

54.6 Email, PDF and Accessibility

- Test schedules, retry, duplicate prevention, safe links, report delivery and escalation.
- Render and visually inspect official PDFs, QR verification and exports.
- Test responsive layouts, keyboard use, focus, labels, contrast and error states.

54.7 Performance, Migration and Recovery

Test realistic data volumes, peak workflows, interrupted jobs, safe retries, upgrade rehearsals, record reconciliation and isolated backup restoration.

54.8 End-to-End and UAT

Execute admission-to-enrolment, teaching-to-results, billing-to-receipt, promotion and management-accountability scenarios. Representative users complete role scripts and record acceptance.

54.9 Final Quality Gate

Release is blocked by build failure, data-loss risk, critical security issues, incorrect calculations, permission leakage, duplicate finance, broken publication, failed migration, unrestorable backup or unresolved UAT blocker.

Chapter Completion Control

This chapter is complete only when its implementation evidence, data effects, permissions, tests, migration or rollback information and unresolved risks have been reported honestly.

CHAPTER 55

55. Deployment, Operational Handover and Final Acceptance

CODEX EXECUTION PROMPT

Apply the Volume IV master rules. Inspect the existing implementation before changing it, preserve all working behavior and historical records, complete only this chapter's current subprompt, run proportionate verification, report evidence and stop.

55.1 Production Readiness

Verify the quality-gate decision, release candidate, migrations, security, performance, recovery, UAT, documentation, training, support and every required production dependency.

55.2 Environment and Release

- Separate development, test, staging and production databases, storage, secrets, queues and providers.
- Enable HTTPS, secure cookies, safe errors, restricted administration, log redaction and startup configuration validation.
- Package the approved build, matching migrations, release manifest, notes and rollback instructions.

55.3 Backup and Migration

- Create and restore-test the pre-deployment database, documents, configuration and current application.
- Restrict access, run migrations, reconcile all principal record totals and stop on material loss or corruption.
- Preserve identifiers, approved results and financial transactions.

55.4 Deployment and Smoke Tests

Activate application, workers, schedules, storage, PDF, email, monitoring and health checks in a controlled order. Verify login, dashboards, student search, class lists, results, finance, files and audit before opening access.

55.5 Institutional Configuration

Configure Ghana and school defaults, academic structure, assessment, grading, promotion, attendance, fees, templates, reports, KPIs, escalation, backup and notifications through approved interfaces.

55.6 Provisioning and Import

Create unique named accounts, scoped roles and secure activation. Import approved operational data using dry runs, validation, duplicate detection, batch audit and reconciliation.

55.7 Documentation and Training

Deliver administrator and role guides, support procedures, practical role training, cut-over plan, fallback process and launch approval.

55.8 Monitoring and Support

Monitor application, database, storage, workers, schedules, email, documents, backups, integrations, errors and suspicious activity. Define support levels, ownership, severity, response and escalation.

55.9 Rollback and Hypercare

Document authorized rollback triggers and recovery sequence. During launch hypercare, review health, defects, permissions, calculations, email, documents and support volume daily.

55.10 Handover

Transfer release information, configuration, operations, security, backup, support and maintenance responsibility.
Transfer secrets through a secure channel, never the general handover document.

55.11 Final Acceptance

Use Fully Accepted, Accepted with Conditions or Rejected Pending Correction. Record deployed version, sign-off roles, remaining conditions, owners, deadlines and verification methods.

55.12 Completion

The enterprise upgrade is complete only when existing grading works, history is preserved, permissions are verified, academic and finance processes reconcile, reports and escalations operate, backups restore, users are trained and institutional acceptance is recorded.

Chapter Completion Control

This chapter is complete only when its implementation evidence, data effects, permissions, tests, migration or rollback information and unresolved risks have been reported honestly.

Volume IV Execution Matrix

Chapter	Execution area	Primary outcome
46	Master control	Safe one-prompt-at-a-time execution
47	Audit	Evidence-based upgrade baseline
48	Backlog	Ordered dependencies and completion rules
49	Database	Preserved and reconciled history
50	Backend	Authorized domain services and APIs
51	Frontend	Role-based accessible experience
52	Automation	Reports, KPIs, email and escalation
53	Security	Verified privacy and record isolation
54	Quality	Traceable testing and release decision
55	Deployment	Controlled launch, handover and acceptance

Final Codex Execution Sequence

Inspect existing system -> establish recoverable baseline -> order the backlog -> migrate data safely -> implement backend services -> implement role experiences -> activate automation -> harden security -> test and reconcile -> deploy -> train -> hand over -> obtain institutional acceptance.

Final Acceptance Statement

Volume IV is approved when every prompt is executable against the existing project without requiring destructive assumptions; completion claims require implementation and test evidence; historical academic and financial records remain protected; and deployment, rollback, support and institutional sign-off are fully defined.

End of Volume IV